Unit 9 Overview: Two- and Three-Dimensional Shapes

Unit Narrative

In this unit, students will use previous knowledge from 6th and 7th grade on area, volume, and surface area to find the areas, volumes, and surface areas of various figures. Students will also discover what effects changing dimensions will have on the volume of a three-dimensional figure and the area of a two-dimensional figure.

The beginning of Section 1 reviews the concept of area with students, which will be used in deriving formulas for volume and surface area. Lesson 2 will have students discover the relationship of population density and area. Lesson 3 in this section where students discover cross sections, which can be used throughout the rest of Unit 9. Students will also discover solids of rotation in Section 1. Section 1 emphasizes area, cross sections, and solids of rotation.

Section 2 extends students' conceptual understanding from previous grades in finding volume. Lesson 5 has students discover Cavalieri's principle, which states that if two three-dimensional figures have bases with the same area and the same heights, the volumes are equivalent. This builds a foundation for finding volume using mathematical and real-world problems.

Section 3 reintroduces surface area, and students derive the formulas and solve mathematical and real-world problems. Lessons 10 and 11 deal with scale factor and dimension changes, and students discover the effects that dimensional changes and scale factor have on the area, volume, and surface area of a figure.

Consider what differentiation might look or sound like for this unit, specifically taking into consideration that some students may have not seen area, volume, and surface area since 6th and 7th grade. Consider providing visuals for students when working with area, volume, and surface area.

Unit At-A-Glance

Section 1: Area and Cross Sections	Benchmarks Addressed	Lesson 1	Finding the Area of Two-Dimensional Figures
	MA.912.GR.4.1 **MA.912.GR.4.2** **MA.912.GR.4.4**	Lesson 2	Density Based on Area
		Lesson 3	Cross Sections
		Lesson 4	Solids of Rotation
Section 2: Volume	Benchmarks Addressed	Lesson 5	Cavalieri's Principle
	MA.912.GR.4.5	Lesson 6	Finding Volume
		Lesson 7	Volume in the Real World
Section 3: Surface Area and Scale Factor	Benchmarks Addressed	Lesson 8	Surface Area of Three-Dimensional Figures
	MA.912.GR.4.3 **MA.912.GR.4.6**	Lesson 9	Surface Area in the Real World
		Lesson 10	Determining How Dilations Affect Area
		Lesson 11	Determining Surface Area Given Dimension Changes

Differentiation	
Universal Design for Learning™ ©CAST, 2008	
Engagement *The "WHY" of learning*	• Bring tangible objects to class to help students have more visuals in Lessons 5-9. See teacher guide for specific lessons for additional ideas or supports. • Most lessons in Unit 9 provide students the opportunity to notice and wonder. Have students move around the classroom to share their notices and wonders with classmates.
Representation *The "WHAT" of learning*	• Have students use colored pencils/pens in every lesson in the unit to mark diagrams, figures, and problems. • Consider providing Play-Dough for students in Lesson 3, and allow students to create the three-dimensional shapes and use dental floss to find the cross sections.
Action & Expression *The "HOW" of learning*	• Allow students to express findings in activities and collaborations through the use of chart or regular paper. There are several lessons in this unit that will allow students to creatively express their knowledge. • Consider having students form expert groups on particular problems in the unit.

ELL Information

For students whose first language is not English, vocabulary assistance should be offered for keywords throughout the topic. Encourage students to use a dual-language dictionary or technology to translate or look up the meaning of the words. Provide students with a graphic organizer, such as a Frayer Model. Offer visual examples of dividing an irregular shape into more familiar shapes, such as triangles and rectangles.

Allow students to work in partners or triads, when appropriate; carefully plan small groups so that ELL students are partnered with non-ELL students. Provide frequent opportunities for students to turn and talk; pair up ELL students with non-ELL students. Encourage students to make dual-language flashcards of the vocabulary words, properties, and theorems, including diagrams where appropriate.

When working with volume, create a new visual aid, chart paper size, and interactive notebook size with a three-column chart/table for the formulas for solids. Label the columns with the NAME of the solid, a SKETCH of the solid, including dimensions, and the FORMULA for the solid. Use real-world examples, such as paper towel rolls for cylinders and tissue boxes for rectangular prisms, and provide students opportunities to interact with geometric solids.

In word problems and on diagrams, have students highlight important vocabulary words, such as height, base, shape of the base, slant height, diameter, radius, etc., and circle important numbers/measurements. In word problems and on diagrams, have students use colored pencils to dissect the figures into more familiar shapes, highlight important vocabulary words, and circle important numbers and measurements.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Area
- Circle
- Composite figure
- Cone
- Cube
- Cylinder (circular)
- Diameter
- Dilation
- Parallelogram
- Pi (π)
- Prism (right)
- Pyramid (regular)
- Radius
- Rectangle
- Rectangular prism
- Rhombus
- Scale factor
- Square
- Sphere
- Trapezoid
- Triangular prism

Additional Differentiated Support

Scaffolding Support

Mark-Up Text can be used to help students approach more complicated and wordy problems as well as provide visual cues for additional feedback between students and teachers.

Teacher will use a word problem where students will be able to identify which parts would need to be represented in numbers and how to write those numbers in an expression or equation.

- Teacher will provide students with the following symbols or students can create their own key for markings.
  - ✓ - background knowledge; they know what to do.
  - ? - confusing information; is this needed to solve the problem?
  - ! - marks where a variable is needed to represent a number.
  - + - extra info in word problem; not needed.
- Using a small piece of text, Teacher will model for students how you would mark up before releasing them to do the same.

On-Ramp

Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish.

- Perimeter, Area, and Volume: Which Should You Use?
- Calculating Volume
- Calculating Surface Area with Nets

Section Overview

Section 1: Area and Cross-Sections (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.912.GR.4.1 Identify the shapes of two-dimensional cross-sections of three-dimensional figures. • *Clarification 1: Instruction includes the use of manipulatives and models to visualize cross-sections.* • *Clarification 2: Instruction focuses on cross-sections of right cylinders, right prisms, right pyramids and right cones that are parallel or perpendicular to the base.* **MA.912.GR.4.2** Identify three-dimensional objects generated by rotations of two-dimensional figures. • *Clarification 1: The axis of rotation must be within the same plane but outside of the given two-dimensional figure.* **MA.912.GR.4.4** Solve mathematical and real-world problems involving the area of two-dimensional figures. • *Clarification 1: Instruction includes concepts of population density based on area.*	Lesson 1	• I can solve mathematical and real-world problems involving the area of two-dimensional figures.
	Lesson 2	• I can solve real-world problems that involve density using area.
	Lesson 3	• I can identify the shapes of two-dimensional cross sections of three-dimensional figures.
	Lesson 4	• I can identify three-dimensional objects by rotations of two-dimensional figures.

Section 2: Volume (Lessons 5-7)

Benchmarks Addressed	Learning Targets	
MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres. • *Clarification 1: Instruction includes concepts of density based on volume.* • *Clarification 2: Instruction includes using Cavalieri's principle to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms and cones.*	Lesson 5	• I can use Cavalieri's principle to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones.
	Lesson 6	• I can find the volume of spheres, prisms, cones, pyramids, and cylinders.
	Lesson 7	• I can use volume to solve problems with real-world applications problems.

Section 3: Surface Area and Scale Factor (Lessons 8-11)

Benchmarks Addressed	Learning Targets	
MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures. **MA.912.GR.4.6** Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.	Lesson 8	• I can solve mathematical problems involving the surface areas of cylinders, pyramids, prisms, cones, and spheres. • I can solve mathematical problems involving the lateral areas of cylinders, pyramids, prisms, cones, and spheres.
	Lesson 9	• I can solve real-world problems involving the surface areas of cylinders, pyramids, prisms, cones, and spheres. • I can solve real-world problems involving the surface areas of composite figures.
	Lesson 10	• I can discover the relationship between area and the scale factor of a dilation.
	Lesson 11	• I can find the area of a figure when one or both dimensions are changed. • I can find the surface area of a three-dimensional figure when one or both dimensions are changed.

Vertical Alignment

7th Grade Unit 12 Surface Area and Volume	**Geometry Unit 9** **Area, Volume, and Surface Area**
Prior Course(s)	**Course Level**
MA.6.GR.2.2 Solve mathematical and real-world problems involving the areas of quadrilaterals and composite figures by decomposing them into triangles or rectangles. **MA.7.GR.1.1** Apply formulas to find the areas of trapezoids, parallelograms, and rhombi. **MA.7.GR.1.4** Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems. **MA.6.GR.2.3** Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. **MA.7.GR.2.3** Solve mathematical and real-world problems involving volume of right circular cylinders. **MA.7.GR.2.1** Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.	**MA.912.GR.4.1** Identify the shapes of two-dimensional cross sections of three-dimensional figures. **MA.912.GR.4.2** Identify three-dimensional objects generated by rotations of two-dimensional figures. **MA.912.GR.4.4** Solve mathematical and real-world problems involving the area of two-dimensional figures. **MA.912.GR.4.5** Solve mathematical and real-world problems involving the volumes of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres. **MA.912.GR.4.6** Solve mathematical and real-world problems involving the surface areas of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres. **MA.912.GR.4.3** Determine how changes in dimensions affect the areas of two-dimensional figures and the surface areas or volumes of three-dimensional figures.

9.1 Finding the Area of Two-Dimensional Figures

Lesson Introduction

Benchmark	MA.912.GR.4.4 Solve mathematical and real-world problems involving the areas of two-dimensional figures.	Learning Target	• I can solve mathematical and real-world problems involving the area of two-dimensional figures.

Benchmark Coherence

Building On	Working Toward
MA.6.GR.2.2 Solve mathematical and real-world problems involving the areas of quadrilaterals and composite figures by decomposing them into triangles or rectangles. MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi. MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.	MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or areas of polygons. MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the areas of two-dimensional figures and the surface areas or volumes of three-dimensional figures. MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface areas of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.

Essential Questions

- What is area?
- How can certain quadrilaterals be divided into triangles and rectangles?
- What are the formulas for area of a rectangle and area of a triangle?

Lesson Narrative

This lesson gives students an opportunity to review areas of triangles and rectangles. Students are given time to explore how to find the areas of various quadrilaterals by dividing the polygons into areas of triangles and rectangles. Students will then use an example of using the area of a circle to find the area of a semicircle. Students can derive formulas for a parallelogram, trapezoid, and rhombus by dividing the figures into rectangles and triangles.

Component Summary

9.1.1 Warm-Up (~5 min.)	Examining a quote from Michael Jordan for growth mindset *(SE p. 51)*	9.1.3 Guided Practice (5-10 min.)	Finding the areas of figures *(SE p. 56)*
9.1.2 Collaboration (20-25 min.)	Exploring the areas of various polygons by decomposing *(SE p. 52)*	9.1.4 Your Turn! (5 min.)	Finding the area of a composite figure *(SE p. 58)*
9.1.5 Wrap-Up (~5 min)	Reflecting on the level of understanding of finding the areas of various figures *(SE p. 58)*		

Resources

Glossary	Materials	Additional Preparation
Key Terms: N/A Additional Terms: N/A	• Calculator • (Optional) Colored Pencils/Pens	• N/A

Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle differentiating between the radius and diameter of a circle. • Students may misread the graphs in question 6 of 9.1.2 Collaboration and question 1 of 9.1.4 Your Turn! as the scale is in units of 0.5 instead of 1. • Students may not notice that in question 1 of 9.1.4 Your Turn! each 1 unit $= 5$ feet. • Students may forget to divide by 2 when finding the area of a triangle.	• Consider allowing students to use tools, such as colored pencils and/or pens to divide the given quadrilaterals into rectangles and triangles so that the visuals might be more accessible for some students.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Discussion Supports 9.1.2 Collaboration lends itself to discussion about the formulas for areas of various figures. Allow students the opportunity to discuss the relationships between the formulas and the figures of the two-dimensional figures and give opportunities for students to construct their own knowledge using discussion supports.	**MA.K12.MTR.5.1: Patterns and Structure** 9.1.2 Collaboration gives students the opportunity to explore and discover how using the areas of rectangles and triangles can be applied to find the areas of other two-dimensional figures. Use MA.K12.MTR.5.1 as an opportunity for students to look at patterns and structures to be able to find the area of multiple two-dimensional figures.

Differentiate Instruction

In 9.1.2 Collaboration, there are opportunities to differentiate with Discussion Supports. The following examples of sentence stems and frames could be used:

Sentence Stems:

- One difference between the formula for the area of a rectangle and the area of a triangle is…
- I can divide the parallelogram into two triangles by…
- I can find the area of a rhombus by…

Sentence Frames: Sentence frames allow all students an opportunity to approach the discussions. The underlined words are the words that would be provided for students to fill in the blanks. See the examples below:

- The radius is half of the diameter.
- The formula for area of a triangle is one half of the base multiplied by the height.
- The formula for area of a rectangle is the base multiplied by the height.

Lesson Notes:

9.1.1 Warm-Up: Michael Jordan

Component Overview: *Students begin the lesson by analyzing a quote from Michael Jordan about how failure leads to success. This gives students an opportunity to think about instances where failures can be turned into successes and how students can grow from failures.*

9.1.1 Warm-Up: Michael Jordan

Basketball legend Michael Jordan is often quoted as saying, "I've failed over and over and over again in my life and that is why I succeed."

1. What is a skill, activity, or subject that you have failed at?

   Answers will vary.

2. Explain Jordan's quote in your own words.

   Answers will vary.

3. How can you turn a failure into a success story?

   Answers will vary.

If time permits, prompt one to two students to share their responses on how failures lead to success. Also, give students an opportunity to describe their interpretations of the quote or offer extra credit for an extension project in which students write a response on how failure leads to success in Math class.

9.1.2 Collaboration: Finding Area

Component Overview: *Students will begin working with the formulas for the area of a triangle and the area of a rectangle, which most students should be able to recall. Students will work with the formulas for the areas of a triangle and rectangle to derive the formulas for the areas a parallelogram, square, trapezoid, and rhombus. 9.1.2 Collaboration also gives students a quick opportunity to review and work with the formula for the area of a circle. For question 4, students may need some additional support and guidance on how to use the triangle to find the area of the parallelogram.*

9.1.2 Collaboration: Finding Area

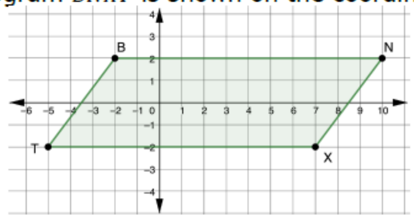
1. Complete the table.

Figure	Area Formula
Rectangle	$A = base \times height$
Triangle	$A = \frac{1}{2} base \times height$

2. How is the area of a triangle related to the area of a rectangle?
Both formulas use the measurements for base and height. The area of a triangle is $\frac{1}{2}$ the area of a rectangle with the same base and height measurements.

Why is the area of a triangle half the area of the rectangle? (Dividing a rectangle diagonally creates two right triangles.)

3. Parallelogram $BNXT$ is shown on the coordinate grid.



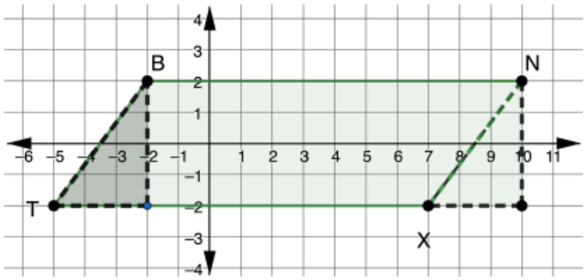
Tyrese could not remember how to the find area of a parallelogram so he divided the parallelogram into 2 triangles and a rectangle. Work with your partner to use Tyrese's method to find the area of $BNXT$.

Shape	Area
Triangle 1	$A = \frac{4 \times 3}{2} = 6\, units^2$
Triangle 2	$A = \frac{4 \times 3}{2} = 6\, units^2$
Rectangle	$A = 4 \times 9 = 36\, units^2$
Total Area of Parallelogram	*The total area of the shape would be the sum of the areas of the individual components. This would be* $36 + 6 + 6 = 48\, units^2$.

How did Tyrese know that dividing the parallelogram into two triangles and a rectangle would work? (The formula for the area of two triangles is equal to the formula for the area of a rectangle.)

9.1.2 Collaboration: Finding Area (continued)

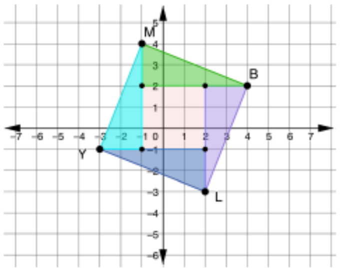
4. Sarianna rearranged the parallelogram to create a rectangle.



Use Sarianna's rectangle to find the area of $BNXT$.

The side lengths of the rectangle are 4 *and* 12. *Thus the area is* $4 \times 12 = 48\ units^2$.

5. Quadrilateral $MBLY$ is decomposed as shown on the coordinate grid. Find the area of $MBLY$ using the decomposition shown.



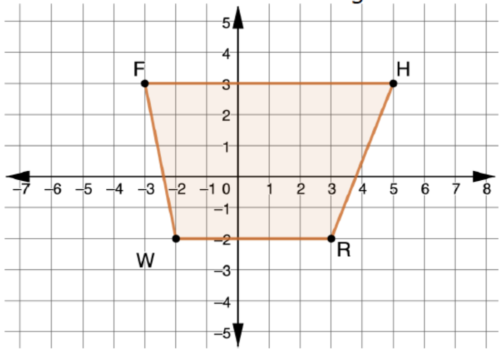
Area of Triangles = $4(\frac{2\times5}{2}) = 20\ units^2$

Square = $3 \times 3 = 9\ units^2$

Total = $9 + 20 = 29\ units^2$

Consider the use of digital applets, such as GeoGebra to visually show the translation of the parallelogram.

6. $FHRW$ is shown on the coordinate grid.



a. Divide the quadrilateral into two triangles and a rectangle. Then, find the area of each.

Shape	Area
Triangle 1	$A = \dfrac{5 \times 1}{2} = 2.5\ units^2$
Triangle 2	$A = \dfrac{5 \times 2}{2} = 5\ units^2$
Rectangle	$5 \times 5 = 25\ units^2$

b. What is the area of $FHRW$?

The area of $FHRW$ is $2.5 + 5 + 25 = 32.5\ units^2$.

c. What type of quadrilateral is $FHRW$?

$FHRW$ is a trapezoid.

7. Ava found a formula for the area of a trapezoid. The formula is $A = \frac{1}{2}h(b_1 + b_2)$, where h is the height and b_1 and b_2 are the bases. Use the formula Ava found to find the area of $FHRW$.

$A = \dfrac{1}{2}(5)(5 + 8) = 32.5\ units^2.$

For a visual approach, have students outline each triangle and the square to be able to better identify each individual figure, especially for students who may have trouble identifying various colors due to vision issues.

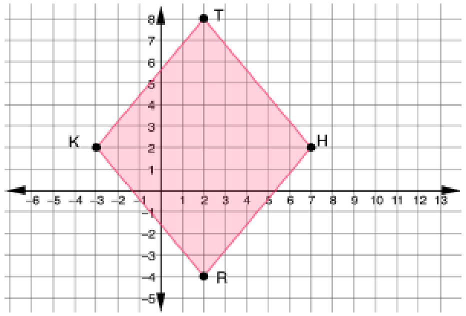
To increase the rigor, have students speculate which specific type of quadrilateral is graphed. Have students justify their answers mathematically. The specifics on identifying quadrilaterals on the coordinate plane will be taught in Unit 13.

For a visual approach, have students use two separate colored pencils or pens to draw the two triangles on the coordinate plane.

How do you know that this quadrilateral is a trapezoid? (The quadrilateral has one set of parallel sides and two slanted sides.)

9.1.2 Collaboration: Finding Area (continued)

8. $THRK$ is shown on the coordinate grid.



a. What type of quadrilateral is $THRK$?

$THRK$ is a rhombus.

b. Determine the area of $THRK$. Show your work or explain your thinking.

Answers will vary. One way to determine the area of a rhombus is the draw diagonals and create 4 congruent right triangles, because diagonals bisect each other and are perpendicular in a rhombus. Each right triangle has an area of $15\ units^2$, so the area of $THRK$ is $60\ units^2$.

c. Complete the table.

Line Segment	Length
$\overline{TR}$	12 units
$\overline{KH}$	10 units

The area of a rhombus can be found using the formula $A = \frac{1}{2}pq$, where p and q are the lengths of the diagonals.

d. Use the formula to find the area of $THRK$.

$$A = \frac{1}{2} \times 12 \times 10 = 60\ units^2$$

Prompt students for verbal explanations and utilize inquiry to extend student thinking for increased rigor.

- *How do you know that this quadrilateral is a rhombus? (The quadrilateral has four congruent sides, perpendicular diagonals, diagonals that bisect each other.)*

Students may categorize quadrilateral $THRK$ as a rhombus. Both answers are correct since, by definition, a rhombus is also a kite.

9.1.2 Collaboration: Finding Area (continued)

Formulas for area can easily be forgotten, so it is always helpful to be able to break a figure down into triangles and rectangles in order to find the area. The exception is finding the area of a circle or a part of a circle.

9. What is the area of a circle?

   $A = \pi r^2$

10. The drama program at school is building a new semicircular stage. The longest distance across the stage is 30 feet. What is the area of the new stage? Round to the nearest thousandth.

$$A = \frac{\pi(15^2)}{2} = 353.429 ft^2$$

Students may need to be reminded of the differences between the radius and diameter. Students may also need to be reminded that if the diameter is given, divide by two to find the radius.

9.1.3 Guided Practice: Using the Formulas

Component Overview: *Students will begin working to find the areas of various two-dimensional figures. Students will find the area of a circle in question 1. Students may divide the figures into triangles and rectangles on questions 2-3, or the formulas for the parallelogram and rhombus that were given in the lesson may be used. Students will be required to divide the figure in question 4 into multiple figures and will be required to find the area of the composite figure.*

9.1.3 Guided Practice: Using the Formulas

1. The radius of a circle is 7 feet (ft). Find the area, in square feet, of the circle. Leave the answer in π form.

   $A = \pi \cdot 7^2 = 49\pi\ ft^2$

2. The base of the parallelogram is 22 inches (in.). The area of the parallelogram is 132 square inches. Find the height, in inches, of the parallelogram.

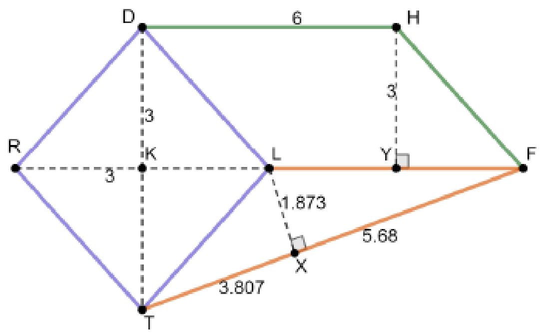
   $A = bh; 132 = 22h; h = 6\ inches$

3. The diagonals of a rhombus are 40 centimeters (cm) and 18 centimeters. Find the area, in square centimeters, of the rhombus.

   $A = \frac{1}{2}(40 \times 18) = 360\ cm^2$

Encourage students to sketch each figure, as students may be hiding underlying misconceptions if they just multiply the numbers in the problem. Sketching each figure provides insight into the students' grasps on concepts like radius vs. diameter (question 1), base vs. height (question 2), and diagonal vs. base (question 3).

4. A composite shape of a rhombus, a parallelogram, and a triangle is shown.



Complete the table. Round to the nearest thousandth.

Shape	Area
Rhombus	$A = \frac{1}{2}pq = \frac{6 \times 6}{2} = 18\ units^2$
Parallelogram	$A = bh = 3 \times 6 = 18\ units^2$
Triangle	$A = \frac{1}{2}bh = \frac{17.77}{2} = 8.885\ units^2$
Composite Area	$18 + 18 + 8.885 = 44.885\ units^2$

The use of colored pencils or markers to clearly annotate the figures that construct the composite figure can be a helpful tool or strategy for students. Encourage students to decompose or mentally deconstruct each figure using the strategy that works best for them.

Students may see the side TL as the base of the triangle, with a height of 6 units, rather than what is depicted in the answer key.

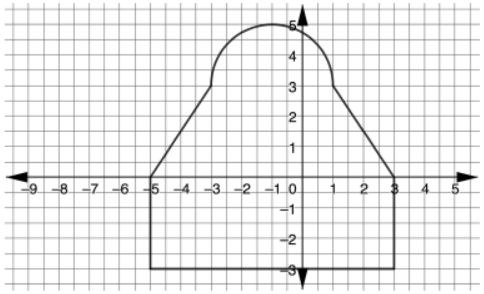
Monitor for students who use 3 units instead of 6 units for the diagonal of the rhombus.

9.1.4 Your Turn!

Component Overview: *Students will complete one question where dividing the composite figure into a semicircle, trapezoid, and rectangle will be required. Students must first complete this step before finding the area of the composite figure. Once the areas of the semicircle, trapezoid, and rectangle have been calculated, students must add to find the area of the composite figure.*

9.1.4 Your Turn!

1. The design of a room is shown, where 1 unit = 5 feet.



Determine the area, in square feet, of the room. Round to the nearest hundredth.

Semi-circle $(r = 2\ units = 10\ ft)$

$= \frac{\pi r^2}{2} = \frac{\pi(10^2)}{2} = \frac{100\pi}{2} = 50\pi = 157.080\ ft^2$

Trapezoid $(h\ =\ 3\ units\ =\ 15\ ft, b_1 = 4\ units = 20\ ft, b_2 = 8\ units = 40\ ft)$

$= \frac{15(20+40)}{2} = \frac{900}{2} = 450\ ft^2$

Rectangle $(b = 8\ units = 40\ ft, h = 3\ units = 15\ ft)$

$= 40ft\ \times 15\ ft = 600ft^2$

Total area $= 157.08 + 450 + 600 =\ 1207.08ft^2$

If students are unsure how to begin, prompt them to use different colors to decompose or "break up" the composite figure into figures that they know.

How do you find the area of a semicircle? (Divide the area of the circle by 2.)

Once students have decomposed the figure into figures they know, prompt students to make connection to what they know about their area.

- *What do you know about the area of a circle?*
- *How is a semicircle related to a circle?*
- *So, what does that make you think about finding the area of a semicircle?*

9.1.5 Wrap-Up: Reflection

Component Overview: *Students complete a self-reflection on their level of understanding of the content of this lesson.*

9.1.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.

I need help. I got it. I could help someone.

Answers will vary.

Finding the area of a triangle	
Finding the area of a rectangle	
Finding the area of a trapezoid	
Finding the area of a square	
Finding the area of a rhombus	
Finding the area of a parallelogram	
Finding the area of a circle	
Finding the area of a composite figure	

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

Mark-Up Text
Consider prompting students to incorporate these symbols throughout this lesson to indicate their level of confidence during the learning process. Students can use one of the three symbols to mark each activity, or you can list each activity on a poster and have students draw their symbol for each activity on the way out of the classroom. Then use these symbols as information to differentiate in Lesson 2.

If time permits, choose one of the strands in the left column and prompt students to find a partner with a different symbol chosen in the right column and collaborate. For example, prompt students who selected "I need help" to find a student who selected "I could help someone" and have them ask that student one question. Students who selected "I got it" can choose to either find someone who needs help or find someone who could help them with a specific question.

9.2 Density Based on Area

Lesson Introduction

Benchmark Focus	MA.912.GR.4.4 Solve mathematical and real-world problems involving the areas of two-dimensional figures.	Learning Target	• I can solve real-world problems that involve density using area.

	Building On	Working Toward
Benchmark Coherence	MA.6.GR.2.2 Solve mathematical and real-world problems involving the areas of quadrilaterals and composite figures by decomposing them into triangles or rectangles. MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi. MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.	MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeters or areas of polygons.

Essential Questions	• What is the relationship between area, population, and population density? • When given the population density, how is the area calculated?
Lesson Narrative	This lesson guides students on finding the population density when given the area and/or the population. Students will engage with real-world data to find the population density when given the area and the population. This lesson also provides several other types of real-world application problems relating to density.

Component Summary	9.2.1 Warm-Up (~5 min.)	Choosing which car you would rather buy *(SE p. 59)*	9.2.3 Guided Instruction (10 min.)	Relating density and area *(SE p. 61)*
	9.2.2 Collaboration (15 min.)	Finding population density *(SE p. 60)*	9.2.4 Guided Practice (10 min.)	Solving contextual problems with density as it relates to area *(SE p. 63)*
	9.2.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 64)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Population Density Additional Glossary: • Area • Density	• (Optional) Colored Pens/Pencils	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may improperly identify the definitions of area and density.
Students may mistakenly identify the area versus the population density when completing questions.
In 9.2.3 Guided Instruction question 3, students may confuse the fractions when converting the units to miles.</td><td>This lesson does not relate the concept of density to the concept of volume, which will be discussed in Lesson 6 of Unit 9.
Consider letting students draw or use colored pens/pencils for the diagrams in the lesson.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Notice and Wonder
In the beginning of 9.2.2 Collaboration, students are given a list of the populations and the areas of the five largest cities by population in Florida. Allow students to do a Notice and Wonder here and gather responses from students as they make the connection between the population and its area.</td><td>MA.K12.MTR.7.1: Real-World Contexts
This lesson is a perfect opportunity to apply mathematics to real-world concepts, as this lesson was written specifically using real-world concepts. Allow students to fully engage with the mathematics in this lesson.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">9.2.2 Collaboration can be differentiated according to students' needs. Visuals would be helpful for students. Instead of having students write the answers in the workbook, have students record the answers on a visual such as a whiteboard, chart paper, or any other vertical non-permanent surface (VNPS). Students will be able to have choice in how their group's information is displayed.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

9.2.1 Warm-Up: Would You Rather?

Component Overview: *Students begin by looking at two scenarios of cars to buy. This gives the students an opportunity to discuss mathematics and to justify their answers using mathematics.*

9.2.1 Warm-Up: Would You Rather?

1. Which car would you rather buy?

Car A	Car B
$19,000 12% depreciation per year when you upgrade the car after 5 years.	$25,000 16% depreciation per year when you upgrade the car after 5 years.

2. Explain your answer.

   Answers will vary. I would rather buy Car B. Even though Car B costs more upfront, it will be worth $10,455.30 after 5 years whereas Car A will only be worth $10,026.91.

This question also provides an opportunity to connect a part of the lesson back to the Literacy and Language Routine for Notice and Wonder. If time allows, have students talk and verbalize their answers.

For an auditory approach to 9.2.1 Warm-Up, allow students to verbally share their answers with either a partner or the whole class.

9.2.2 Collaboration: Population Density

Component Overview: *Students will begin working with the tables shown, where the five largest cities by population are given as well as the area of each city. Through 9.2.2 Collaboration, students will complete a Notice and Wonder as well as collaborate with other classmates to discover the relationship between population and area to determine how to calculate population density. The data for this assignment is real-world data as of March 2021.*

9.2.2 Collaboration: Population Density

1. The five largest cities in Florida and their populations are shown.

City	Population
Jacksonville	890,467
Miami	454,279
Tampa	387,916
Orlando	280,832
St. Petersburg	261,338

The table shows the area of Florida's five largest cities.

City	Area (square miles)
Jacksonville	747
Miami	35.87
Tampa	113.41
Orlando	102.4
St. Petersburg	61.74

What Do You Notice About the Population and the Area?
Answers will vary. Highest population is not necessarily the same thing as population density – the city of Miami squeezes almost half a million people into 36 square miles.

2. Discuss with your partner your understanding of population density. Describe the population density for the city or town you live in.
Answers will vary. I think it means the amount of people per square mile – or how many people you fit into an area.

Is there a specific relationship between area and population?

For extra differentiation, allow students to choose whether to discuss verbally or discuss by writing down their answers in the workbook. See the Differentiate Instruction guidance section on page 2 for other ideas.

Students should be discussing the math here and should be given the opportunity to do so throughout this Collaboration. Be sure to tie in the Notice and Wonder Literacy and Language Routine as well as MA.K12.MTR.4.1.

9.2.2 Collaboration: Population Density (continued)

3. Make predictions about the population density for Florida's five largest cities.
Answers may vary.

Largest Population Density	Smallest Population Density
Miami	*Jacksonville*

4. Explain your thinking.
Answers will vary.
I think Miami has this highest population density – it's fitting so many people into an area that is quite small. I estimate that Jacksonville will have the lowest population density because even though it has the highest population, it also has the largest area by a factor of 7.

5. Ask a classmate for the predictions they made as well as their reasons. Record their cities and write your summary of their reason. *You are the only person who should write in the middle two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

	City	My Summary of Their Reason	Initials
Largest Population Density		*Answers will vary*	
Smallest Population Density		*Answers will vary*	

To find population density, divide the population by the area.

$$\text{Population Density} = \frac{\text{population}}{\text{area}}$$

For a kinesthetic approach, divide the room into five sections (one for each city) and have students share their predictions by going to the city they think has the largest population density. Repeat by doing the smallest population density.

Each corner then chooses a spokesperson for their corner to provide rationale for their response.

Students can then choose to stay in their corner or reshuffle based on their arguments.

9.2.2 Collaboration: Population Density (continued)

6. Determine the population density, in persons per square mile $(\text{persons}/\text{mi}^2)$, of the five largest cities in Florida and record them in the table from largest to smallest. Round answers to the nearest whole number.

City	Population Density (persons/ mi.^{2})
1. *Miami*	12,665
2. *St. Petersburg*	4,233
3. *Tampa*	3,420
4. *Orlando*	2,743
5. *Jacksonville*	1,192

7. On July 1, 2019, Florida had a population of 21,477,737. The area of Florida is 53,624.76 square miles. Find the population density of Florida.
~401 people per square mile

For a kinesthetic approach, have cards listed with the city and the population density and allow students to do a quick card sort after doing the calculations.

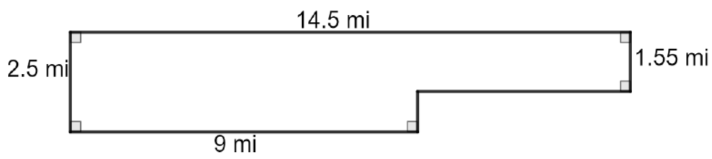
How does the population density of the state of Florida compare to the five largest cities in Florida? (The population density of the state as a whole is smaller than the population density of the five largest cities.)

9.2.3 Guided Instruction: Area and Density

Component Overview: *Students will begin working with contextual problems relating to area and density. Students will find the area and the population density in question 1 and notice a trend. Question 2 also requires students to find area and then use population density to determine the population of panthers in a certain area. Question 3 requires conversions of units to find population density.*

9.2.3 Guided Instruction: Area and Density

1. An outline of the city of Miramar, Florida, with measurements in miles (mi), is shown.



What strategy would be the best to use to find the area of the city of Miramar? (Divide the shape of the city into two separate rectangles and then add the areas together.)

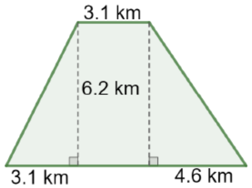
a. What is the area of the city, in square miles?
I made the shape into two rectangles.
Rectangle 1: $A = bh; A = 5.5 \times 1.55 = 8.53\ mi^2$
Rectangle 2: $A = bh; A = 9 \times 2.5 = 22.5\ mi^2$
$22.5 + 8.53 = 31.025\ mi^2$
The area of the city is $31.03 mi^2$

b. Determine the population density of Miramar for 1995 and 2015. Include units.

Year	Population	Population Density
1995	48,099	$1{,}550\ ^{people}/_{mi^2}$
2015	136,996	$4{,}415\ ^{people}/_{mi^2}$

c. How did the population density change between 1995 to 2015?
The population density almost tripled – meaning there are almost three times as many people on the same amount of land.

2. The Florida Fish and Wildlife Conservation Commission (FWC) tracks panthers in Florida. By tracking a panther, FWC can map the area in which a panther lives. The habitat of "Florida Panther 251" is shown with dimensions in kilometers, km.



a. What is the area, in square kilometers (km^2), of Florida Panther 251's habitat?

$A = \frac{1}{2}h(b_1 + b_2) = \frac{1}{2}6.2(3.1 + 10.8) = \frac{1}{2}6.2(13.9) = \frac{1}{2}86.18 = 43.09\ \text{km}^2$

At one time, the entire southeastern United States was the habitat of the Florida panther. Today, their habitat is restricted to several national parks and wildlife preserves in southwest Florida. Today, the total combined area of the Florida panther habitat is $9{,}466\ km^2$.

b. Based on the area of Florida Panther 251's habitat, how many Florida panthers could be living in southwest Florida?

$9{,}466 \div 43.09 = 219.67$, *because you cannot have* $\frac{2}{3}$ *of a panther, I will round up. So* 220 *panthers could be living in southwest Florida.*

Students may calculate question 2a by either using the formula for the area of a trapezoid or by decomposing into rectangles and triangles and then adding together.

Monitor students on question 2b. Students may try to calculate population density instead of population. Also, students must round up their answers to the nearest whole number.

9.2.3 Guided Instruction: Area and Density (continued)

3. Town A has 20 city blocks, and each city block measures $\frac{1}{20}$ of a mile by $\frac{1}{2}$ mile. There are $6{,}500$ people in this town. Find the population density of the town.

Area of 1 block: $A = bh = \frac{1}{20} \times \frac{1}{2} = \frac{1}{40} mi^2$

Area of city blocks in Town A:
$A = \left(\frac{1}{40} \times 20\right) = \frac{1}{2} mi^2$

Population density: $\frac{6{,}500}{\frac{1}{2}} = 13{,}000$ *people / mi*^{2}

What strategy would be the best to use to find the area of the town?

Consider having students draw a picture here to help better understand how to find the area of the town.

Provide additional scaffolding with students as needed when working with fractions to find the length and width of the town. Monitor that students are multiplying the fractions by 20.

9.2.4 Guided Practice: Density Based on Area

Component Overview: *Students will begin working with the population, area, and population density in more contextual questions. The problems in 9.2.4 Guided Practice are more complex than the other problems in the lesson. In question 2, students are given the population density and the persons per square mile, and it asks students to find the area of the city.*

9.2.4 Guided Practice: Density Based on Area

1. The number of leaves on a tree can be estimated using the crown of the tree. A tree crown is the top part of the tree where branches grow out from the main trunk. The width of the tree's crown is used as the diameter of the circle of the ground that lies under the tree's crown. The area where the leaves could fall is 4 times the area of the circle that lays under the tree's crown. The approximate number of leaves per square foot is based on the average size of a leaf for each type of tree. Use the table to approximate the number of leaves that could fall from each tree.

Tree	Crown (in feet)	Approximate Number of Leaves per Square Foot
Sycamore	35	5
Live Oak	16	14

Sycamore

Area of crown:

$$A_{crown} = \pi(17.5)^2 = 962.11$$

Area of leaves:

$$A_{leaves} = 4(962.11) = 3848.44$$

Number of Leaves:

$$5 = \frac{\#\ of\ leaves}{3848.44}$$

$$\#\ of\ leaves = 19242.2$$

Oak

Area of crown:

$$A_{crown} = \pi(8)^2 = 201.06$$

Area of leaves:

$$A_{leaves} = 4(201.06) = 804.24$$

Number of Leaves:

$$14 = \frac{\#\ of\ leaves}{804.24}$$

$$\#\ of\ leaves = 11259.36$$

2. The population density of Gainesville, Florida, is 2028.4 persons per square mile, and the population of Gainesville is $124{,}504$. What is the area, in square miles, of Gainesville?

Population density $= \frac{population}{area}$, *so area* $= \frac{population}{population\ density}$

using the terms given, this gives us an area of $61.38\ mi^2$.

Consider having students draw the two circles to represent the sycamore and live oak trees to have a visual representation of the question.

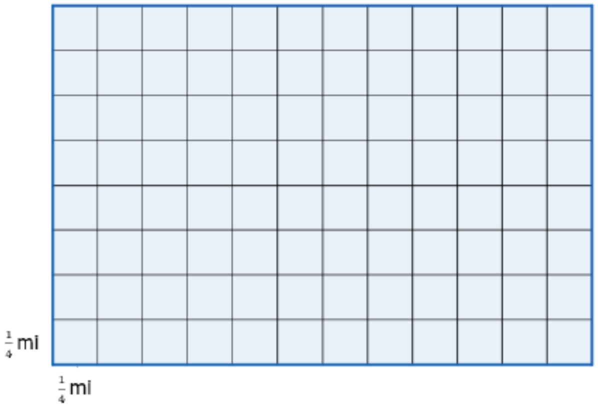
Once students have drawn visuals, prompt student thinking by asking the following questions below:

- *Does crown represent the diameter or the radius of the circle? (Diameter.)*
- *How do you find the radius of a circle when given diameter? (Divide by 2.)*
- *How do you find the number of leaves after you have found the area? (Multiply by the number of leaves per square foot.)*

Ask students how to find the area when given the population and people per square mile.

9.2.4 Guided Practice: Density Based on Area (continued)

3. Wildlife biologists divided a habitat in the Bear Island area of the Big Cypress National Preserve into 96 squares, as shown. Each square has the dimensions of $\frac{1}{4}$ mile (mi) by $\frac{1}{4}$ mi.



The biologists found that in the Bear Island habitat, there are 13.8 deer per square mile.

a. What is the population density for one of the $\frac{1}{4}$ mi by $\frac{1}{4}$ mi square?

There would be 16 of the $\frac{1}{4}$ mile squares within one square mile. Therefore, we would divide $13.8 \div 16 = 0.86$ deer per square $\frac{1}{4}$ mile.

b. How many deer live in the habitat area the biologists studied?

There are a total of 6 mi² and 13.8 deer per square mile. So, $6 \times 13.8 = 82.8 \approx 83$ deer in the entire space.

Consider having students use colored pencils and/or pens to divide and label each section on the grid.

9.2.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

9.2.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about population density.

   Answers will vary.

Consider giving some extra support with the sentence stems given. Encourage students to be as specific as possible here for self-reflection.

9.3 Cross Sections

Lesson Introduction

Benchmark Focus	MA.912.GR.4.1 Identify the shapes of two-dimensional cross-sections of three-dimensional figures.	Learning Target	• I can identify the shapes of two-dimensional cross sections of three-dimensional figures.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What are cross sections? • How can you determine the shape of the cross section of a three-dimensional figure? • How do you sketch the shape of a cross section of a three-dimensional figure?
Lesson Narrative	This lesson gives students a collaboration opportunity to determine the shape of cross sections of three-dimensional figures. Students make connections between the three-dimensional figures and their cross sections. The latter part of the lesson makes students identify and sketch the cross section. The lesson has a mix of collaboration as well as guided and independent work.

Component Summary				
	9.3.1 Warm-Up (~5 min.)	Finding a pattern in a sequence of images *(SE p. 65)*	9.3.3 Guided Instruction (15-20 min,)	Drawing the cross sections of figures on a grid *(SE p. 67)*
	9.3.2 Collaboration (10-15 min.)	Determining the shape of cross sections of prisms, cylinders, cones, and spheres *(SE p. 66)*	9.3.4 Your Turn! (5 min.)	Practice drawing cross sections of figures *(SE p. 69)*
	9.3.5 Wrap-Up (~5 min.)	Drawing the cross sections of figures on a grid *(SE p. 70)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms</u>: N/A <u>Additional Glossary:</u> • Cross Section	• (Optional) Colored Pens/Pencils • (Optional) Straightedge	• Consider using manipulatives and videos that provide a visual demonstration of cross sections.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>• Students may incorrectly identify cross sections of a horizontal plane when compared to a vertical plane.
• Students may not correctly apply the dimensions properly when sketching cross sections.</td><td>• Consider providing additional help and/or scaffolding on 9.3.3 Guided Instruction questions 3 and 4, as these questions require specific dimensions to accurately draw the cross sections.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Three Reads
The questions in the 9.3.3 Guided Instruction and 9.3.4 Your Turn! give a perfect opportunity to execute the Three Reads strategy. Have students read the problems three times to be sure they fully comprehend what the questions are asking. Here is an example of three reads.

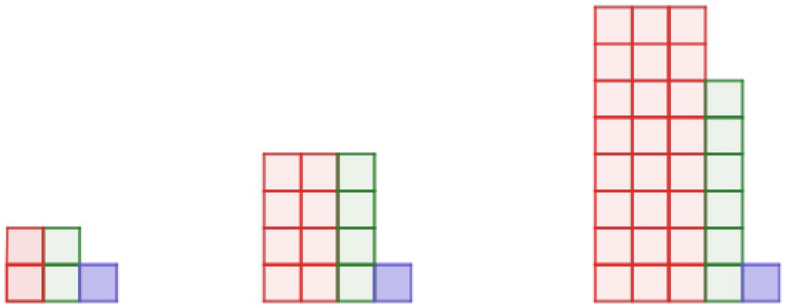
Read 1: Teacher reads the question aloud. Ask students to consider: What is this situation about?
Read 2: Class does choral read or partner read of the question stem. Ask students to consider: What are the quantities in the situation?
Read 3: Partner or choral read the problem stem orally one more time. Ask students to consider: What mathematical questions can we ask about the situation?</td><td>MA.K12.MTR.2.1: Multiple Representations
9.3.1 Warm-Up provides an opportunity to use multiple representations when describing a sequence. Students may draw all seven shapes, make a table of values, or write an algebraic equation to represent the sequence. Provide an opportunity for students to share their representations and discuss how each representation means the same thing.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The concept of cross sections is visual in nature. Use this lesson to differentiate for students who tend to think visually. Some students may struggle with conceptualizing cross sections. For a visual entry-point, use manipulatives and videos that demonstrate cross sections.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

9.3.1 Warm-Up: Patterns

Component Overview: *Students are given a diagram and are asked to use patterns to determine the number of red squares in step 7.*

9.3.1 Warm-Up: Patterns

1. How many red squares are in the seventh pattern?



Answers will vary. The horizontal dimension of the red blocks is n, where n is the pattern number. The vertical dimension of the red blocks is 2^n. The total number of red squares for any pattern number, n, is $n(2^n)$. For the seventh pattern there will be $7(2^7) = 896$ red squares.

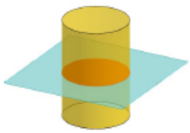
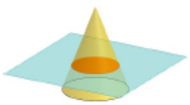
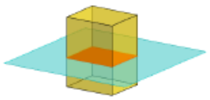
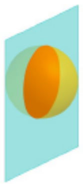
To highlight MA.K12.MTR.2.1 (Multiple Representations), some students could make a table of values to find the solution. Some students may also be persuaded to use a formula. Others may choose to draw all seven images. Allow students multiple pathways to find a solution to this problem. Provide an opportunity for students to share their representations and then make connections between them.

9.3.2 Collaboration: Cross Sections

Component Overview: *Students will examine the table of five three-dimensional figures and their vertical and horizontal cross sections. A Notice and Wonder will be completed to guide students through the process of learning about cross sections and how to identify the cross section of a three-dimensional figure. Students will complete a table with the three-dimensional figures by identifying the cross sections.*

9.3.2 Collaboration: Cross Sections

The table below shows the resulting cross section for each type of slice.

Figure	Slice	
	Horizontal Plane	Vertical Plane
Cylinder		
Cone		
Rectangular Prism		
Triangular Prism		
Sphere		

Use the following questions as prompts to guide student thinking and understanding of how cross sections are formed:

- *Why are the horizontal and vertical cross sections of the cylinder, cone, and triangular prism different?*
- *Why are the horizontal and vertical cross sections of the sphere and the rectangular prism the same?*

9.3.2 Collaboration: Cross Sections (continued)

1. What do you notice about the diagrams for horizontal slices?
   Answers will vary. I notice that the horizontal diagrams match the base of the three-dimensional object.

2. What do you notice about the diagrams for vertical slices?
   Answers will vary. I notice that the vertical diagrams make shapes that you might not see from looking at the 3-D shape.

3. Discuss with your partner whether or not the shape of a cross section will change if the height of the plane is changed. Summarize your discussion.
   Answers will vary. In vertical slices, the size changes but not the overall shape because the height of the figure is one of the slice's dimensions. In horizontal slices, the shape, nor size, of the cross section will not change when the height of the figure changes.

4. Complete the table by describing the shape of the cross section that is created when the figure is sliced by each.

Figure	**Cross Section Shape**	
	Horizontal Plane	**Vertical Plane**
Cylinder	*Circle*	*Rectangle*
Cone	*Circle*	*Triangle*
Rectangular Prism	*Rectangle*	*Rectangle*
Triangular Prism	*Triangle*	*Rectangle*
Sphere	*Circle*	*Circle*

The Notice and Wonder provides students with a discussion opportunity. Have students jot down their Notice and Wonder and allow students to discuss their observations verbally.

For a kinesthetic approach, allow students to move to other students in the classroom and give an opportunity to discuss and share their noticings and wonderings.

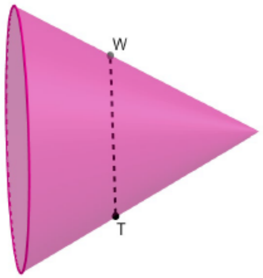
9.3.3 Guided Instruction: Drawing Cross Sections

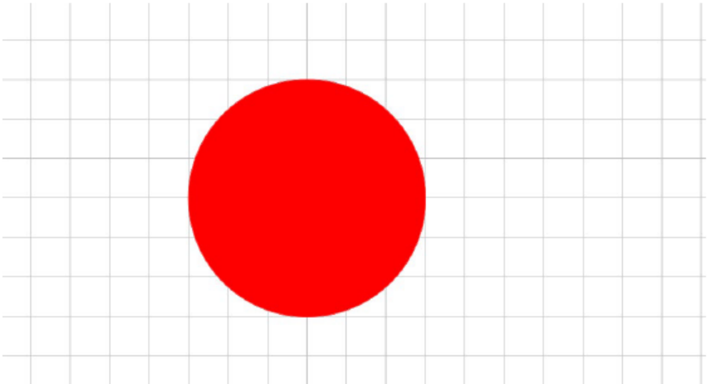
Component Overview: *This component of the lesson extends from 9.3.2 Collaboration by having students sketch the resulting cross sections on a grid when given the three-dimensional figure. Each question gives dimensions that students are expected to apply to sketch the cross section.*

9.3.3 Guided Instruction: Drawing Cross Sections

Three-dimensional figures can be oriented in different ways. Cross sections are typically described by their relation to the base of the three-dimensional figure. For example, cross sections are often described as parallel or perpendicular to the base of the figure.

1. A cone is shown with a cross section parallel to the base. Draw the cross section of the cone on the grid, given $WT = 6$ units.

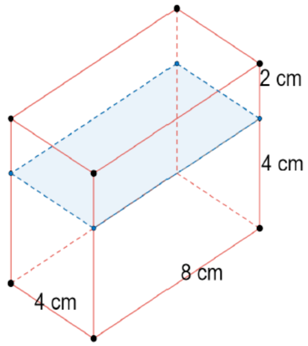


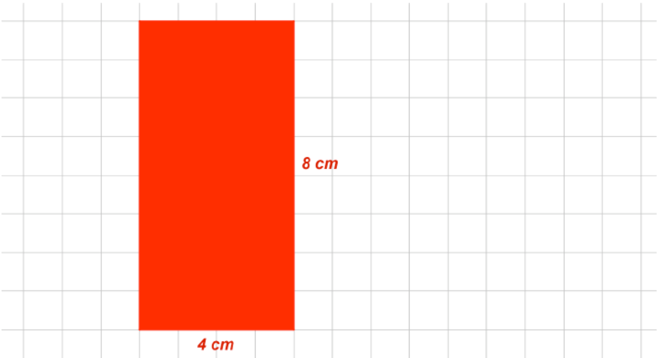


How would the vertical cross section change if the cone was standing on its base instead of its side?

9.3.3 Guided Instruction: Drawing Cross Sections (continued)

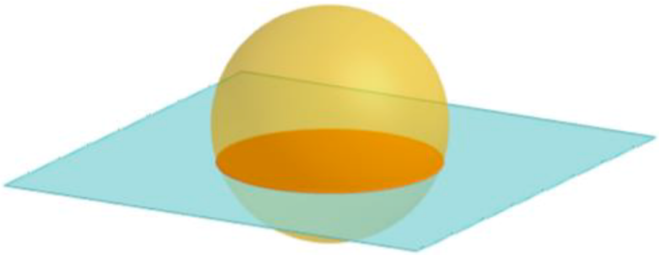
2. A rectangular prism is shown with a horizontal cross section. The dimensions are given in centimeters (cm). Draw the cross section on the grid and label its dimensions.



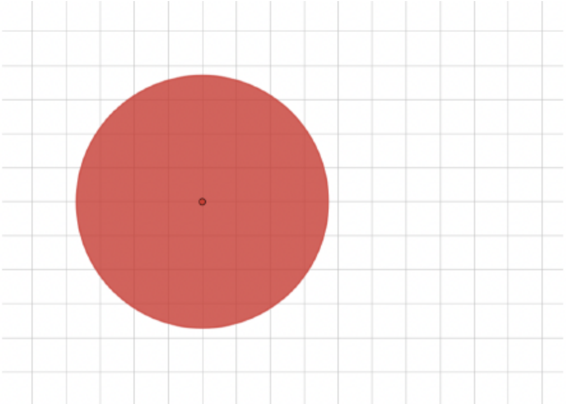


Be sure that students label the proper dimensions when graphing the cross sections to indicate what one unit of the grid represents.

3. A sphere with radius of 4 inches (in.) with a horizontal cross section is shown.



Draw the cross section if it occurs $\frac{3}{8}$ of the distance from the sphere's center.



A cross section that is $\frac{3}{8}$ of the distance from the center of the sphere will be located $4 \cdot \frac{3}{8} = \frac{6}{4} = 1.5$ in. from the center of the sphere. Using Pythagorean Theorem, $1.5^2 + b^2 = 4^2$, the cross-section circle that is $\frac{3}{8}$ of the distance from the center of the sphere will have a radius of a distance of $\sqrt{13.75} \approx 3.708$ inches.

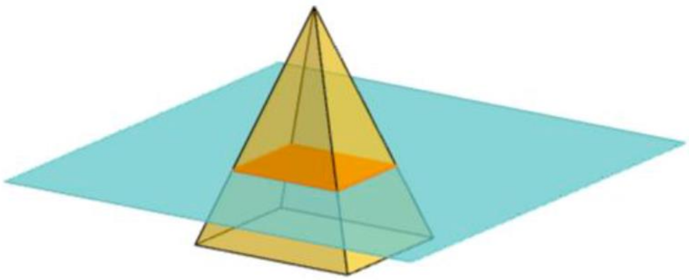
In the answer key, one unit of the grid represents 1 inch.

Students may require additional help and scaffolding for question 3. Consider asking the following when needed:

- *If the cross section is $\frac{3}{8}$ of the distance from the center, how far is it from the edge of the sphere?*
- *How can you find the radius of the cross-section circle that is scaled from the center of the sphere? (Multiply 4 by $\frac{5}{8}$.)*

9.3.3 Guided Instruction: Drawing Cross Sections (continued)

4. A rectangular pyramid is shown, where the dimensions of the base are 12 centimeters (cm) by 21 cm and the pyramid's height is 33 cm.



Draw the cross section if the horizontal plane is located at a height of 11 cm.

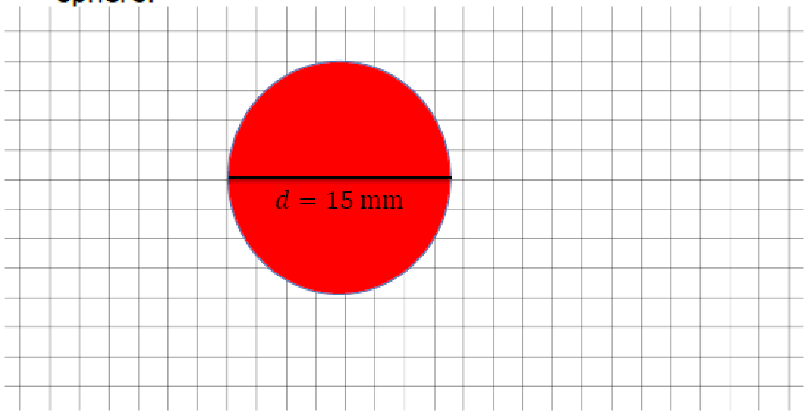
Provide opportunity for additional scaffolding as needed on question 4. Students will have to multiply the dimensions by $\frac{2}{3}$ because the cross section is $\frac{2}{3}$ the height of the pyramid (the perpendicular distance between the top of the pyramid to the bottom).

9.3.4 Your Turn!

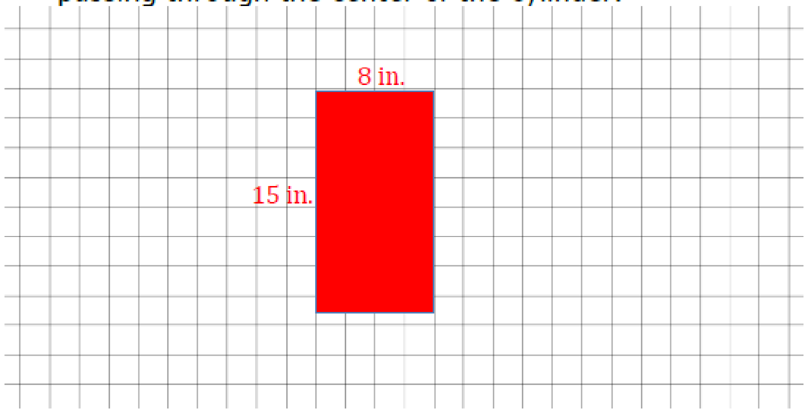
Component Overview: *Students practice drawing two-dimensional cross sections of three-dimensional figures on a grid.*

9.3.4 Your Turn!

1. Draw the cross section formed by the intersection of a sphere with a diameter of 15 millimeters and a perpendicular plane passing through the center of the sphere.



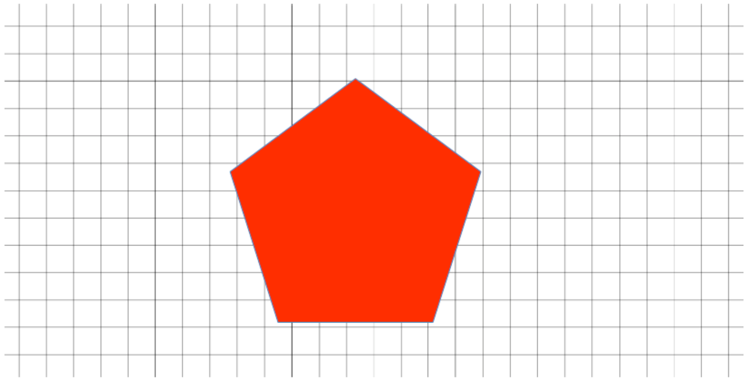
2. A cylinder has a radius of 4 inches (in.) and a height of 15 in. Draw the cross section formed by the intersection of a cylinder and a plane perpendicular to the base passing through the center of the cylinder.



Before sketching, have students identify what shape a cross section is of the intersection of a sphere and a perpendicular plane. For further scaffolding, review the term "perpendicular." Also, have students pay attention to the dimensions given so that the cross sections are sketched accurately. In the shown answer key for questions 1 and 2, one unit of the grid represents 2 millimeters. Allow students to make choices of what one unit represents for their drawings.

9.3.4 Your Turn!

3. Draw the cross section formed by the intersection of a pentagonal prism and a horizontal plane passing through the prism.



For differentiation, assign students one of the three questions and have a student become an "expert" on the question. Then, group students into groups of three, where each student was assigned a different question. Then have students share how to complete their questions.

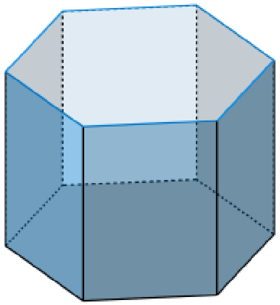
Question 3 has no given dimensions of the pentagonal prism. Correct answers will be any-size pentagon.

9.3.5 Wrap-Up: Ticket Out the Door

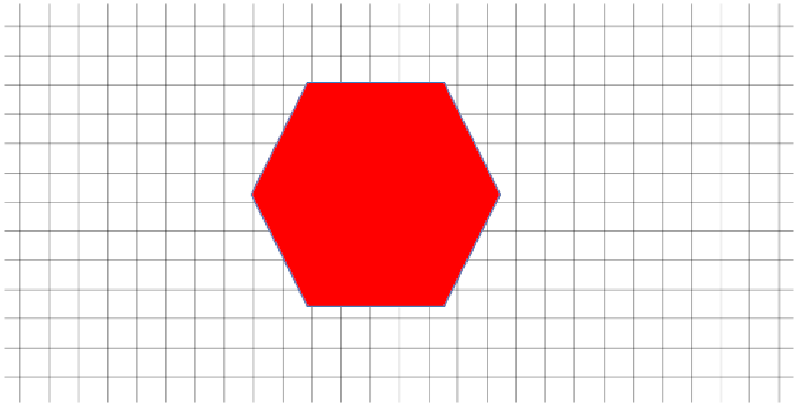
Component Overview: *Students complete a question where they are required to sketch a cross section that is horizontal to the plane and vertical to the plane.*

9.3.5 Wrap-Up: Ticket Out the Door

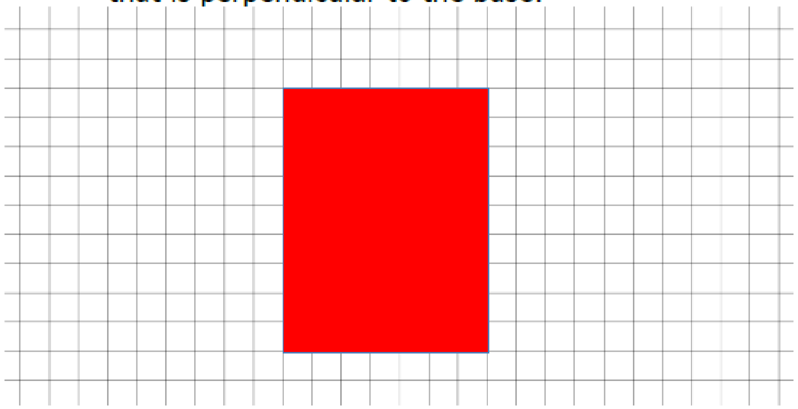
1. A hexagonal prism is shown.



   a. Draw the cross section that results from a plane that is parallel to the base.



   b. Draw the cross section that results from a plane that is perpendicular to the base.



Cross sections are important for Lesson 4 in Unit 9. Consider using the results of 9.3.5 Wrap-Up to form groups for 9.4.2 Activity.

Questions 1a and 1b have no given dimensions of the three-dimensional shapes. Correct answers will be any-size hexagon and rectangle. The shape of the rectangle will vary depending on where the plane passes through the shape.

9.4 Solids of Rotation

Lesson Introduction

Benchmark	MA.912.GR.4.2 Identify three-dimensional objects generated by rotations of two-dimensional figures.	Learning Target	• I can identify three-dimensional objects generated by rotations of two-dimensional figures.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions

- How can you identify three-dimensional objects generated by rotations of two-dimensional figures?
- How can you identify the two-dimensional figure that was used to generate a three-dimensional object by rotation?

Lesson Narrative

This lesson allows students to use cross sections and has students perform rotations to see what three-dimensional objects would be formed by a rotation around a given axis. Students discover the three-dimensional objects formed by various two-dimensional figures. The lesson also has students work backward by giving three-dimensional objects, and students have to determine what two-dimensional figure would have formed the three-dimensional object by rotation.

Component Summary

9.4.1 Warm-Up (~5 min.)	Sketching cross sections of a Rubik's Cube and a Slinky *(SE p. 71)*	9.4.3 Guided Instruction (10 min.)	Sketching two-dimensional figures or three-dimensional objects that create or result from rotations *(SE p. 74)*
9.4.2 Activity (15-20 min.)	Describing and sketching three-dimensional objects created by rotating two-dimensional figures *(SE p. 71)*	9.4.4 Guided Practice (5-10 min.)	Describing and sketching three-dimensional objects created by rotating two-dimensional figures *(SE p. 75)*
9.4.5 Wrap-Up (~5 min.)	Describing and sketching three-dimensional objects created by rotating two-dimensional figures *(SE p. 75)*		

Resources

Glossary	Materials	Additional Preparation
Key Terms: N/A Additional Terms: N/A	• Sticky Notes • Pencils • (Optional) Graphing Utility • (Optional) Honeycomb Paper Decorations • (Optional) Drill and Cardboard Rectangle, Triangle, and Semicircle	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may mistake the vertical axis and the horizontal axis.
Students may incorrectly graph two-dimensional figures and three-dimensional objects due to not using the correct dimensions, such as height and radius.</td><td>Consider using honeycomb paper decorations to model solids of rotation.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Notice and Wonder
This lesson provides a plethora of opportunities for students to notice and wonder, especially in 9.4.1 Warm-Up and 9.4.2 Activity. The lesson opens with a Notice and Wonder, and although not specifically stated in the 9.4.2 Activity, students are required to make predictions, which requires students to notice and wonder.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
This lesson is discussion-heavy, especially throughout 9.4.2 Activity. Students are required to determine which three-dimensional object would be formed through a rotation and how that would be sketched. Students will also analyze the mathematical thinking of other students throughout this lesson.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider providing visuals and other manipulatives to create kinesthetic entry-points for students. Using visuals can help all students better visualize how solids of rotation work. For example, digital applets like GeoGebra’s Solids of Rotation offer students the chance to digitally explore the rotation two-dimensional figures and the three-dimensional objects that they form.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

9.4.1 Warm-Up: Notice and Wonder

Component Overview: *Students complete a Notice and Wonder given a picture of a Rubik's Cube and Slinky. Students are also expected to sketch the horizontal cross section of the cube and the Slinky.*

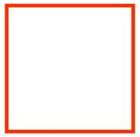
9.4.1 Warm-Up: Notice and Wonder

A Rubik's Cube and a Slinky are shown.

1. What do you notice?
   Answers will vary. They both could form patterns.

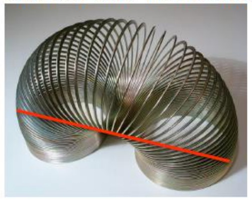
2. What do you wonder?
   Answers will vary. How could they be turned or stretch to make new patterns?

3. Sketch the horizontal cross section of the Rubrik's cube.



The red lines are on picture will form a square.

4. Sketch the horizontal cross section of the Slinky.



Answer will vary.

Allow students to share their notices and wonders from questions 1 and 2 prior to sketching the cross sections in questions 3 and 4.

For question 4, students may interpret the cross section differently. Some may see the cross section as the shape of an "athletic track" while others may see an oval. If time permits, entertain a quick discussion of what they see.

Consider asking the following questions:
- *What would the horizontal cross section of the Slinky be if it were in a stacked position and not stretched over? (Circle.)*
- *What would the vertical cross section of the Slinky be if it were in a stacked position and not stretched over? (Rectangle.)*

9.4.2 Activity: Rotating Pencils

Component Overview: *9.4.2 Activity has students working with partners to discover solids of rotations by rotating pencils. Students will be given sticky notes (or a 3x5 index card) to tape to the side of a pencil. Students will answer guided questions to discover which three-dimensional objects will be formed with rotations.*

9.4.2 Activity: Rotating Pencils

Work with your partner to complete the following.

1. Place a sticky note on the side of the pencil. If necessary, use tape to secure the sticky note.

2. Make a prediction about which three-dimensional shape the sticky note would form when the pencil is rotated $360°$.

   Complete the table.

	Prediction	Reasoning
Sticky Note	*Answers will vary. cylinder*	*Answers will vary. Going around the pencil makes a circle. With it being a pencil to hold and a 3-dimensional shape, it is in the form of a cylinder.*

3. Using your pencil as a vertical axis of rotation, rotate the pencil $360°$.

4. What three-dimensional figure did the sticky note seem to create?
   cylinder

Throughout 9.4.2 Activity, students will complete questions by rotating their pencils and noticing and wondering to make their predictions and observations.

9.4.2 Activity: Rotating Pencils (continued)

5. Work with your partner to predict what three-dimensional shape will be formed when each shape is rotated $360°$.

Shape	Prediction	Reasoning
Rectangle	*Cylinder*	*Answers will vary.* *As a rectangle rotates around it forms a circular shape of a cylinder.*
Triangle	*Cone*	*Answers will vary.* *When it is rotated, it forms a circular base with triangular sides.*
Semicircle	*Sphere*	*Answers will vary.* *Half of a sphere is a hemisphere since a semicircle is half of a circle, but when it rotates a sphere will form.*

6. Use sticky notes or paper to create a rectangle, a triangle, and a semicircle to attach to your pencil. Using your pencil as a vertical axis of rotation, rotate the pencil $360°$. Then, sketch or describe the resulting three-dimensional shape.

Shape	Three-Dimensional Shape
Rectangle	*Cylinder. It's like folding a rectangle around a pencil, and it taking the shape of the pencil.*
Triangle	*Cone. As the triangle rotates around the pencil, it's like forming an ice cream cone.*
Semicircle	*Hemisphere. If the vertical axis is the axis of symmetry, it will make a hemisphere, resembling a the top of a muffin.*

Consider opportunities to connect student predictions to visual representations such as the honeycomb paper decorations or applets, like GeoGebra's Solids of Rotations.

Consider allowing students to share their responses before having students rotate their pencils to see if students' predictions match the resulting three-dimensional objects.

Direct the students to create each shape using sticky notes. The teacher can also use cardboard or other thick board to cut out shapes and demonstrate the rotations using a drill.

9.4.2 Activity: Rotating Pencils (continued)

7. Discuss with your partner whether the three-dimensional shape would be different if the pencil were held horizontally instead of vertically. Summarize your discussion.
   Answers will vary. The figures formed by rotation could remain the same if the axis of rotation is in the same location relative to the 2D object. However, if the 2D shape keeps its orientation and the axis of rotation (pencil) is horizontal, then the 3D shape that is formed will change. The rectangle makes a cylinder with different dimensions. The triangle will vary depending on the orientation of the triangle and location of the axis. The semicircle makes a sphere.

Students may describe different three-dimensional objects depending on the type of triangle they rotated. Also, where the figure was taped to the pencil will influence the object formed. Ask probing questions about these differences while students share out their findings.

8. Draw any four-sided, two-dimensional shape on a 3x5 card. Make it as unique as possible. Cut out your shape. Draw the outline of your shape below.

   Answers will vary.

9. Sketch what you think the three-dimensional figure of your four-sided two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis
Answers will vary.	*Answers will vary.*

The teacher may also use a digital approach on questions 8 and 9 if necessary.

10. Review your partner's shape. Then, complete the following.
    - Draw your partner's shape on the grid below.
    - Draw a horizontal axis of rotation.
    - Draw the reflection of your partner's shape.

Answers will vary.

9.4.2 Activity: Rotating Pencils (continued)

11. Discuss with your partner how drawing a reflection could help envision the three-dimensional figure. Summarize your discussion.

    Answers will vary. A reflection on the other side of the axis will help to see what the direction the shape is going as well as how the shape is folding around the axis asked.

12. Sketch what you think the three-dimensional figure of your partner's four-sided, two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis
Answers will vary.	*Answers will vary.*

13. Compare your sketches with your partner. Discuss any differences and make any adjustments.

    Answers will vary.

Allow students the opportunity to notice and wonder when working with partners on question 10.

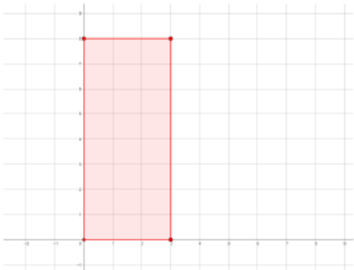
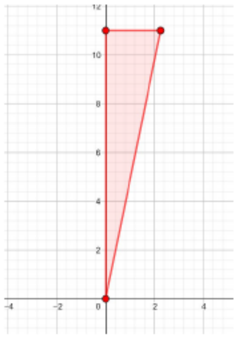
Questions 12-13 give students the opportunity to look at multiple representations given the horizontal and vertical axes.

9.4.3 Guided Instruction: Rotations With Dimensions

Component Overview: *Students will sketch two-dimensional and three-dimensional figures and must work with both representations. In question 1, an image of a three-dimensional object is given, and students must sketch the two-dimensional figure that would be formed. Question 2 requires students to sketch a three-dimensional object when given a two-dimensional figure and an axis of rotation.*

9.4.3 Guided Instruction: Rotations with Dimensions

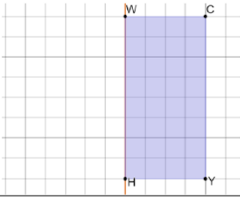
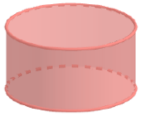
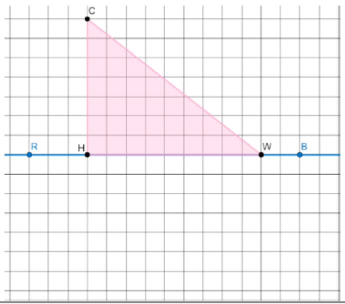
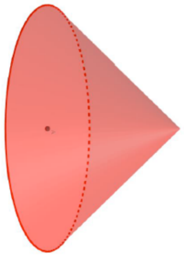
1. Draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

Image	Two-Dimensional Figure
A set of revolving doors with a radius of 3 feet. 	
An ice cream cone with a diameter of 4.5 centimeters (cm) and a height of 11 cm. 	

Have students who are struggling look at the pictures to help determine what two-dimensional figure is formed. Engage students in conversation about what the various cross sections of the object would look like at various locations and what part of the cross section is needed to create the object. Monitor to see if students are graphing correctly.

9.4.3 Guided Instruction: Rotations With Dimensions (continued)

2. Sketch the three-dimensional shape that is created. Include any dimensions in units.

Two-Dimensional Shape	Three-Dimensional Shape
$WCYH$ is rotated around $\overleftrightarrow{WH}$. 	 *Cylinder* *Radius: 4 units* *Height: 8 units*
ΔCWH is rotated around $\overleftrightarrow{RB}$. 	 *Cone* *Radius: 7 units* *Height: 9 units*

How can students find the dimensions of the three-dimensional objects using the two-dimensional figures?

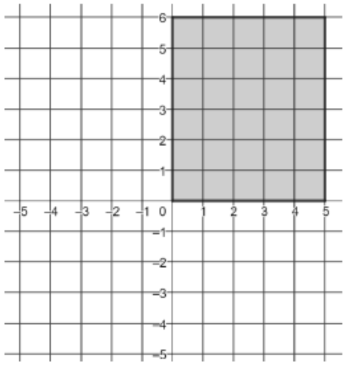
What are the similarities and differences between rotating the triangle around $\overleftrightarrow{RB}$ and rotating it around $\overline{HC}$? (Similarities are that they both create a cone shape; differences are the height and radius of the two cones will be different.)

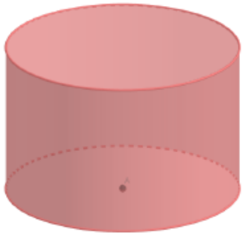
9.4.4 Guided Practice: Rotations

Component Overview: *Students complete two questions where students sketch a three-dimensional object when given a two-dimensional figure on the coordinate plane.*

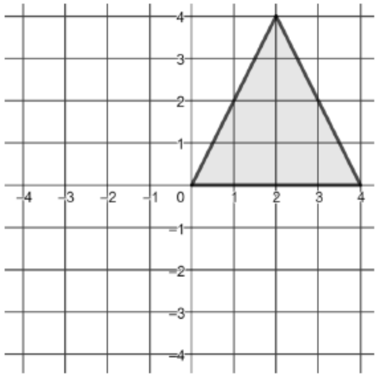
9.4.4 Guided Practice: Rotations

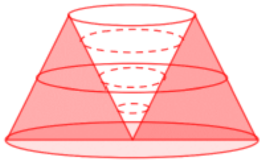
1. Describe and sketch the three-dimensional shape that is created when the rectangle is rotated around the y-axis. Include any known dimensions.
   Cylinder with radius of 5 units and a height of 6 units.





2. Describe and sketch the three-dimensional shape that is created when the isosceles triangle is rotated around the y-axis. Include any known dimensions.
   Answers will vary. Truncated cone of height 4 and a radius of 4 with a cone of height 4 and radius of 2 removed from the inside.





For students who struggle with drawing, consider making a "bank" of figures for the students to choose from that represents what the rotated image will look like. Pay closer attention to the students' descriptions than to their abilities to represent the figure through a drawing.

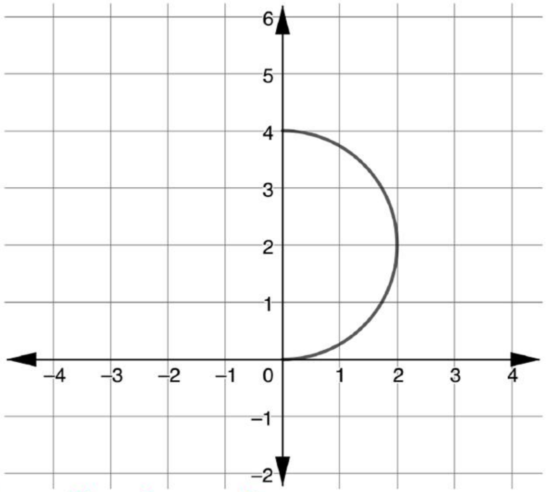
For students who finish early, challenge them to create a third two-dimensional figure on the coordinate plane and draw the resulting three-dimensional object that is created when rotated around an axis.

9.4.5 Wrap-Up: Ticket Out the Door

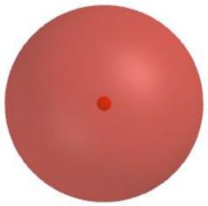
Component Overview: *Students complete a question to measure their understanding of the content of the lesson.*

9.4.5 Wrap-Up: Ticket Out the Door

1. Describe and sketch the three-dimensional shape that is created when the semicircle is rotated around its diameter. Include any known dimensions.



Sphere with radius 2 units.



Consider using student responses from this activity for differentiation in future lessons.

9.5 Cavalieri's Principle

Lesson Introduction

Benchmark	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.	Learning Target	• I can use Cavalieri's principle to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones.

Benchmark Coherence	Building On	Working Toward
	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders. MA.912.GR.4.1 Identify the shapes of two-dimensional cross sections of three-dimensional figures.	N/A

Essential Questions	• How can Cavalieri's principle be used to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones?
Lesson Narrative	Students are guided through the purpose of Cavalieri's principle and how it can be used to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones. The 9.5.2 Collaboration gives students several scenarios for the development of Cavalieri's principle while the rest of the lesson gives students the opportunity to put Cavalieri's principle into practice. This lesson also includes instruction on vertical and slant height.

Component Summary				
	9.5.1 Warm-Up (~5 min.)	Notice and wonder about records that are stacked differently *(SE p. 76)*	9.5.4 Guided Instruction (5-10 min.)	Defining slant height and vertical height *(SE p. 79)*
	9.5.2 Collaboration (5-10 min.)	Exploring scenarios to develop Cavalieri's principle *(SE p. 77)*	9.5.5 Guided Practice (10 min.)	Finding volume using the idea of Cavalieri's principle *(SE p. 80)*
	9.5.3 Guided Instruction (5-10 min)	Formalizing Cavalieri's principle *(SE p. 78)*	9.5.6 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 82)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Cavalieri's Principle for Three-Dimensional Solids • Slant Height • Vertical Height • Altitude <u>Additional Terms:</u> N/A	• (Optional) Tangible Objects for 9.5.5 Guided Practice (i.e. stack of sticky notes, stack of chocolate squares, stack of compact discs, and glass vase)	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>On question 4b in 9.5.5 Guided Practice, students may forget to subtract parts of the vase. See note in teacher guide.</td><td>Consider having tangible visuals for students to access throughout this lesson to help fully solidify the concept of Cavalieri’s principle.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Notice and Wonder
There are multiple chances for students to notice and wonder throughout the lesson. By having real-life scenarios, students can notice that if three-dimensional objects have the same areas of bases and the same heights, then the volumes are congruent. Give students the opportunity to notice and wonder in the lesson.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
Lesson 5 gives ample opportunities for students to discuss and construct their own knowledge. Provide structures and the environment that is conducive for students to engage in discussion of mathematics. Allow students to construct their own arguments with respect to Cavalieri’s principle and the volume of a three-dimensional object.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider providing objects for students to manipulate to show that if the heights of two three-dimensional objects are congruent and the areas of the bases are congruent (even if the shape of the bases are not congruent), then the volumes are congruent. There are several examples throughout the lesson to provide a visual. Teachers may also provide their own examples of visuals to demonstrate Cavalieri’s principle.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

9.5.1 Warm-Up: Notice and Wonder

Component Overview: *Students examine four pictures and record a Notice and Wonder.*

9.5.1 Warm-Up: Notice and Wonder

Four different perspectives of the same stack of records are shown.

1. What do you notice?
   Answers will vary.
   All are viewed from different angles.
   Some perspectives make it seem like there are more or less records that are in the stack.

2. What do you wonder?
   Answers will vary.
   Why are some stacked neatly and others not?
   Why does a change in perspective make it seem like there are more or less records in the stack?

Provide students with the opportunity to share their noticings and wonderings with the class. By having real-life scenarios, students can notice that if three-dimensional objects have the same areas of bases and the same heights, then the volumes are congruent.

9.5.2 Collaboration: Exploring Cavalieri's Principle

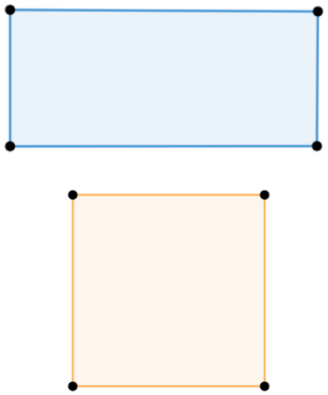
Component Overview: *Students collaborate and are given three distinct scenarios. Scenario 1 asks students to determine the relationship between the volumes of two stacks of pennies. Scenario 2 gives students two towers with the same heights and area but different bases. The last scenario describes the volumes between a cone and a pyramid.*

9.5.2 Collaboration: Exploring Cavalieri's Principle

Scenario 1	
Consider the two stacks of pennies, where each stack contains the same number of pennies. Discuss with your partner the relationship between the volumes of each stack.	

1. Summarize your discussion in the table below and have your partner initial that your summary is correct.

Summary	Partner's Initials
Answers will vary. *Volumes will be the same because the number of pennies is the same.*	

Scenario 2	
Suppose two partners are both constructing towers that are to be 5 inches high. Partner A decided to build a tower using rectangles with areas of 16 square inches and partner B decided to construct a tower using squares with areas of 16 square inches. The base of each partner's tower is shown.	

Discussion provides additional auditory entry-points. Consider utilizing sentence stems and providing ample time for students to grapple with academic language:

- *"I think the volume of each of the stacks of pennies is _______, because the height of each stack of pennies is _____."*
- *"I notice in the picture that _______, which makes me think _____."*

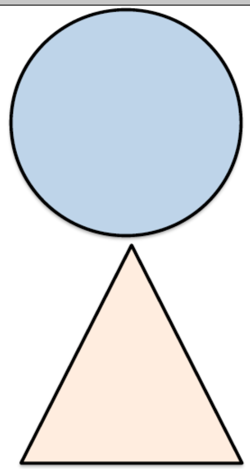
If necessary, provide students with a tangible stack of pennies for students to see the volumes are the same because the heights are the same.

For Scenario 2, consider using a graphing utility that has 3D capabilities.

9.5.2 Collaboration: Exploring Cavalieri's Principle (continued)

2. Discuss with your partner how the volumes will compare. Summarize your findings in the box below and have your partner initial that your summary is correct.

Summary	Partner's Initials
Answers will vary. The volumes will be the same because they each have a height of 5 inches and the area of the bases will be the same.	

Scenario 3	
ABC ice cream shop decided to run a special for the week and they're offering two serving options for you to choose from. Option 1 is a cone with a circular base that has an area of 4π square inches and a height of 4 inches. Option 2 is a triangular pyramid container with a base of 4π square inches and a height of 4 inches. The base of each option is shown.	

3. Discuss with your partner how the volumes will compare to one another. Summarize your findings in the box below and have your partner initial that your summary is correct.

Summary	Partner's Initials
Answers will vary. The volumes will be the same because the area of the bases will be the same.	

Ask the following questions to guide students through 9.5.2 Collaboration:

- *How will the volumes be different if the areas of the bases and the heights are the same?*
- *Does the shape of the base affect the total volume?*

Ask the following questions to guide students through Scenario 3:

- *What do you notice about the area of both bases?*
- *What do you notice about the heights of the cone and the pyramid?*

9.5.3 Guided Instruction: Cavalieri's Principle

Component Overview: *Students work through two questions on using Cavalieri's principle. The first question requires students to compare two three-dimensional objects and determine whether the volumes are equal. The second question ties in cross sections and formulas for area and volume.*

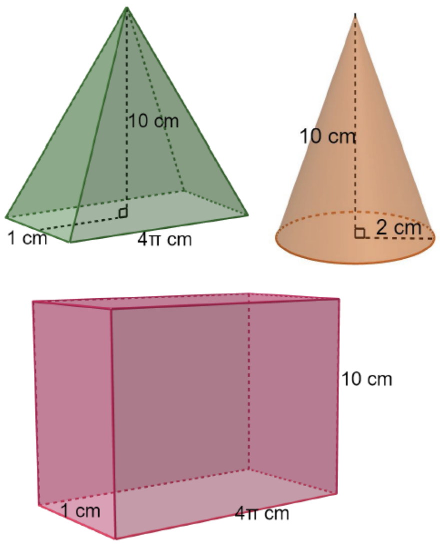
9.5.3 Guided Instruction: Cavalieri's Principle

Cavalieri's Principle for Three-Dimensional Solids

If every plane parallel to the two bases of two solids results in cross sections of equal area and the two solids have congruent altitudes, then the solids have equal volumes.

What information do you need to know to determine whether the solids have the same volume?

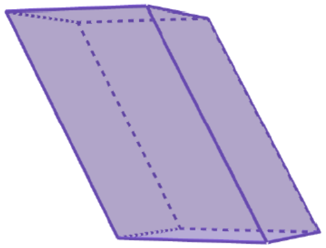
1. Using Cavalieri's Principle, determine whether or not the following solids have the same volumes.



No. A cross section that is 1 cm from the base of the pyramid will not have an area that is the same as a cross section that is 1 cm from the base of the rectangle.

9.5.3 Guided Instruction: Cavalieri's Principle (continued)

2. An oblique prism is shown.



a. What shape is each cross section of the oblique prism?
Parallelogram

b. What is the area formula for the base?
$A = lw$

c. What is the volume formula for an oblique rectangular prism?
$V = lwh$

d. Discuss with your partner how the formula for the oblique rectangular prism compares to the formula for a right rectangular prism. Explain your reasoning.
Answers may vary. The formulas are the same just based off if the shape is two dimensional or three dimensional. The shape is designed different than normal.

How do you determine the shape of the cross section of the oblique prism shown?

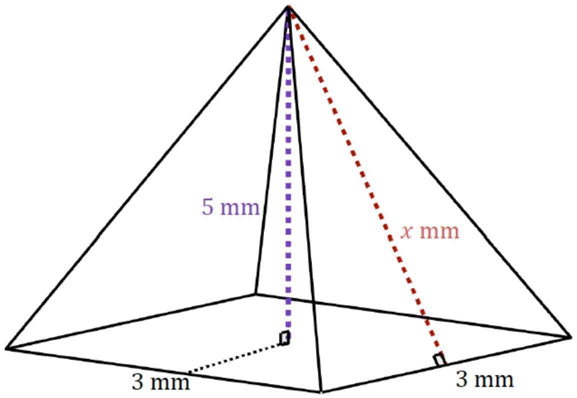
Allow students the opportunity to share out their responses to question 2d. Use this as an opportunity to address misconceptions that arise during 9.5.3 Guided Instruction.

9.5.4 Guided Instruction: Slant Height vs. Vertical Height

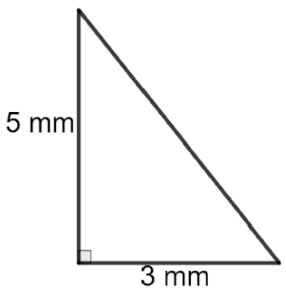
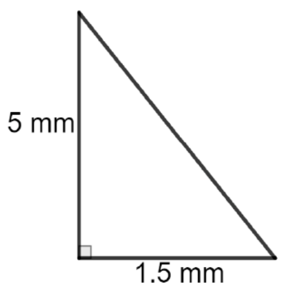
Component Overview: *Students are guided through a comparison of slant height and vertical height and how to find the slant height when given the vertical height.*

9.5.4 Guided Instruction: Slant Height vs. Vertical Height

A square pyramid is shown with measurements in millimeters (mm).



Samantha and Julian each found the measure of the slant height, x, by first drawing a diagram and then applying the Pythagorean Theorem. Their work is shown.

Samantha	Julian
	
$c^2 = 3^2 + 5^2$	$c^2 = 1.5^2 + 5^2$

Have students discuss the similarities and differences between vertical height and slant height.

Why is slant height needed instead of vertical height to find the volume of a square pyramid?

9.5.4 Guided Instruction: Slant Height vs. Vertical Height (continued)

1. Discuss with your partner which student correctly set up their work to determine the slant height. Summarize your discussion.

   Julian is correct because he used the correct distance from the base to the center for that leg side of the Pythagorean theorem.

The **slant height** of an object is the distance from the base to the apex along a lateral face, whereas the **vertical height or altitude** of an object is the perpendicular distance from the apex to the base.

2. Discuss with your partner which height is important when finding the volume of an oblique figure. Summarize your discussion.
   Answers may vary. The vertical height or altitude is more important when finding the volume of an oblique figure.

Have a few students share their responses with other partners or classmates from question 1 and then repeat again in question 2.

Slant height is often a concept that students confuse with the traditional vertical height of a triangle. Spend some time discussing with students the impact of choosing the wrong value and why the values can be different.

9.5.5 Guided Practice: Cavalieri's Principle in the Real World

Component Overview: *Students complete several questions relating to Cavalieri's principle. Question 1 discusses a stack of sticky notes. Question 2 gives students a picture of two different stacks of chocolate squares. Question 3 gives students a scenario with compact disks (CDs), and Question 4 gives students a picture of a vase and a context.*

9.5.5 Guided Practice: Cavalieri's Principle in the Real World

1. Consider the stack of sticky note pads below.

If each pad is a square with side lengths of 3 inches and a height of 0.97 inches, approximate the volume of a stack of 8 sticky note pads.

$V = (3)(3)(0.97)$
$V = 8.73$
$V = 8.73(8)$
$V = 69.84 in.^3$

Does the volume change since the stack of sticky notes are not perfectly stacked? Why or why not?

2. Two stacks of 12 chocolate squares are shown.

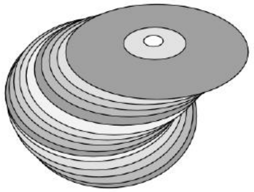
Stack A	Stack B
	

Each chocolate square has the dimensions $1\frac{5}{8}$ inches (in.) by $1\frac{5}{8}$ in. Each square is $\frac{1}{8}$ in. thick. Find the volume of each stack.

Stack A	Stack B
$V = \left(1\frac{5}{8}\right)\left(1\frac{5}{8}\right)(1.5) = 3.96 in.^3$	$V = \left(1\frac{5}{8}\right)\left(1\frac{5}{8}\right)(1.5) = 3.96 in.^3$

What is something that you notice about the two stacks? What is something you wonder about them?

3. While cleaning out an attic, you found an old stack of compact discs (CDs) that were used to store music.



If each disc has an area of 36π cm^2, explain how you can find the approximate volume of a stack of 25 CDs.

Multiply the area of the disc by the total vertical height of the stack of CDs.

4. A zigzag crystal vase is shown, where the vase has a height of 10 inches (in.). The cross-sections that are parallel to the base are always rectangles that are 6 in. wide by 3 in. long.

a. Determine the volume of the vase, assuming the crystal itself has no thickness.

$V = lwh$

$V = (6)(3)(10)$

$V = 180 in.^3$

Consider having tangible visuals for these items as students are working through 9.5.5 Guided Practice. If available, try to bring some blank CDs in for students to have a visualization for question 3 to better connect to the content as they complete the question.

9.5.5 Guided Practice: Cavalieri's Principle in the Real World (continued)

b. The crystal is actually half an inch thick on each of the sides and on the bottom. Approximately how much space is contained within the vase? Explain or show your reasoning.

$6 - 1 = 5$

$3 - 1 = 2$

$10 - \frac{1}{2} = 9\frac{1}{2}$

$V = 5(2)\left(9\frac{1}{2}\right)$

$V = 95 in.^3$

Since each of the sides are 0.5 inches thick, the length and width of the cross-sectional rectangle will each decrease by 1 inch. Meanwhile, height of the vase itself will decrease by 0.5 inch due to the thickness of the glass.

Help students through question 4b if necessary. Some students may forget to subtract the $\frac{1}{2}$ inch thickness on each side of the vase, thus resulting in incorrect calculations. Also, students will only subtract $\frac{1}{2}$ inch on the bottom.

9.5.6 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

9.5.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
   Answers will vary.

Have students share their responses with a partner if time allows.

9.6 Finding Volume

Lesson Introduction

Benchmark	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.	Learning Target	• I can find the volume of spheres, prisms, cones, pyramids, and cylinders.

	Building On	Working Toward
Benchmark Coherence	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volumes of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. MA.7.GR.2.3 Solve mathematical and real-world problems involving volumes of right circular cylinders.	N/A

Essential Questions	• How can I find the volume of spheres, prisms, cones, pyramids, and cylinders? • What are the formulas for the volume of spheres, prisms, cones, pyramids, and cylinders? • How can I derive the formulas for the volume of spheres, prisms, cones, pyramids, and cylinders?
Lesson Narrative	This lesson guides students to develop a conceptual understanding of the formulas for finding volume. The collaboration in 9.6.2 Guided Instruction provides students the opportunity to discuss the mathematics and discover the relationships between the volumes of the cylinder, cone, pyramid, rectangular prism, and the sphere. This lesson also introduces the concept of composite figures and how to find their volumes.

Component Summary	9.6.1 Warm-Up (~5 min.)	Choosing between two pools based on dimensions *(SE p. 83)*	9.6.3 Guided Practice (5-10 min.)	Finding the volume of a hemisphere and a composite shape *(SE p. 87)*
	9.6.2 Guided Instruction (15-20 min.)	Finding volume of spheres, prisms, cones, pyramids, cylinders, and composite figures *(SE p. 83)*	9.6.4 Your Turn! (5-10 min.)	Finding the volume of a cylinder and a pyramid *(SE p. 87)*
	9.6.5 Wrap-Up (~5 min.)	Finding the volume of a composite figure *(SE p. 88)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Cone • Pyramid • Sphere <u>Additional Terms:</u> • Volume • Prism • Cylinder	• (Optional) Water • (Optional) Rice, Sugar, or Sand	• If available, find a 3D model of a sphere, prism, cone, pyramid, and cylinder.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may divide the radius by 2 instead of dividing the volume by 2 to find the volume of a hemisphere. • Students may use the wrong formulas to find the volume of the individual three-dimensional figures (i.e. students may confuse the volume formulas for cylinders and cones). • Students may substitute in the diameter measure instead of the radius measure into a formula. • Students may not find the individual volumes of the figures that make up a composite figure.	• Consider having students make a formula chart to reference the formulas after students have made connections and derived the formulas. • If 3D models are not available, consider using an online visual such as GeoGebra to model the three-dimensional figures.
Literacy and Language Routines	**Mathematical Thinking and Reasoning (MTR) Standard**
Notice and Wonder The 9.6.2 Guided Instruction discovery portion of the lesson lends itself well to noticing and wondering, especially relating to the formulas for volume. Allow students the opportunity to notice and wonder throughout the lesson, as this should better help students conceptualize the formulas for volume.	**MA.K12.MTR.6.1: Reasonableness** Students will work through a series of questions to anticipate how the volume of a cone relates to a cylinder and the volume of a pyramid relates to a prism. Students then use the formulas to determine the volumes of figures. Students should evaluate their results, asking, “Does this solution make sense?” to monitor the reasonableness of their solutions.

Differentiate Instruction

Consider using visuals for students to conceptualize the concept of volume. There are many examples of applets on GeoGebra that would help students to better understand how volume formulas are derived. This will provide another entry-point for students to derive the formulas for the volumes of various three-dimensional figures.

Lesson Notes:

9.6.1 Warm-Up: Would You Rather?

Component Overview: *Students begin by deciding which of the two pools they would want to have. Students are to justify their answers using mathematics.*

9.6.1 Warm-Up: Would You Rather?

1. Would you rather have a pool that is 20 meters (m) by 8 m by 6 m, or a pool that is 1550 centimeters (cm) by 650 cm by 725 cm?

   a. Use mathematics to compare the two pools.

   Pool 1 volume: $20 \times 8 \times 6 = 960\ m^3$

   Pool 2 volume: $15.5 \times 6.5 \times 7.25 = 730\ m^3$

   b. Which pool would you rather have?

   Answers will vary. I would rather have pool 1 as it is larger.

To effectively compare the sizes of the two pools, students must convert centimeters to meters.

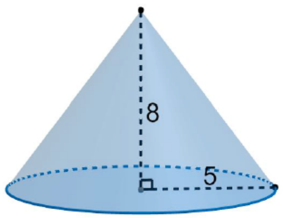
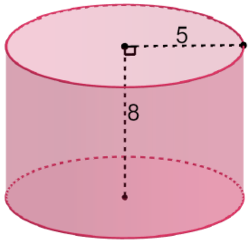
For a kinesthetic approach for 9.6.1 Warm-Up, poll the class and see how many students chose which option. Polling could involve a simple thumbs-up.

9.6.2 Guided Instruction: Finding Volume

Component Overview: *Students will begin working with the sphere, cone, cylinder, prism, and pyramid and derive formulas for finding the volume of each. The purpose of this component is to conceptualize the volume formulas by showing the similarities between the cone and the cylinder as well as similarities between the rectangular prism and the pyramid. Composite figures are also briefly introduced.*

9.6.2 Guided Instruction: Finding Volume

1. A cone and cylinder are shown, where the radius of the base of each is 5 units and the height of each is 8 units.

Cone	Cylinder
	

   a. Suppose the cone is filled with water. Discuss with your partner how many cones you think you would need to fill the cylinder with water. Write your estimate.
   Answers will vary. I think that there would be 3 cones worth of water to fill the cylinder.

   b. Given the volume of a cylinder is $V = Bh$, discuss with your partner how to rewrite the formula to find the volume of the cone.

   Answers will vary. I know it would need to involve pi and a fraction – it is smaller than the cylinder.

   c. The volume formula for a cone is $\underline{V = \pi r^2 \frac{h}{3}}$

A ***cone*** is a three-dimensional figure with a circular base and an apex that is connected to the base by a collection of line segments that form a curved surface.

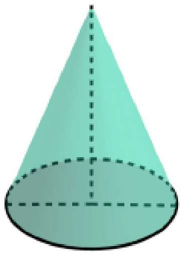
To better prompt student thinking about the formulas of the cone and the cylinder, consider using the following questions:

- *What are the main differences between the cone and the cylinder?*
- *How are the cone and the cylinder similar?*
- *How can the similarities of the cone and cylinder help to develop a formula to find the volume?*

For a kinesthetic approach for this question, consider having a 3D model of a cylinder and of a cone. Pour either water, sugar, sand, or rice from the cone into the cylinder to show that it takes the volume of three cones to make a cylinder.

9.6.2 Guided Instruction: Finding Volume (continued)

2. A cone is shown with a diameter of 16 millimeters (mm), a height of 22 mm, and a slant height of 23.4 mm.



Determine the volume of the cone, in terms of π.

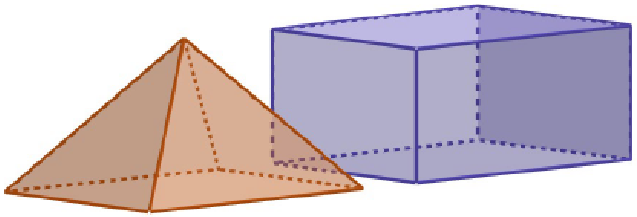
$$V = \pi r^2 \frac{h}{3}$$

$$= \pi 8^2 \frac{22}{3}$$

$$= \pi(64)\left(\frac{22}{3}\right)$$

$$= \frac{1408}{3}\pi\ mm^3$$

3. The bases of the rectangular pyramid and rectangular prism shown have congruent bases, and each has the same height.



a. Suppose the rectangular pyramid is filled with water. Discuss with your partner how many rectangular pyramids you think you would need to fill the rectangular prism with water. Write your estimate.
Answers will vary. I believe that 4 rectangular pyramids would fill the rectangular prism.

For a visual approach to this activity, consider the use of digital applets for making sense of how many pyramids fill a rectangular prism.

What are the main differences between the rectangular prism and the pyramid?

b. Given the volume of a prism is $V = B \cdot h$, discuss with your partner how to rewrite the formula to find the volume of the pyramid.

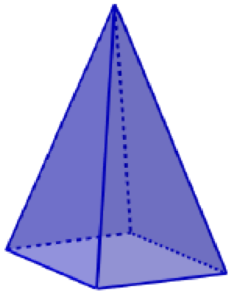
Answers will vary. I believe that the formula might be $v = \frac{1}{2} b \cdot h$

c. The volume formula for a regular pyramid is $V = \dfrac{l \cdot w \cdot h}{3}$

A ***pyramid*** is a figure with a polygonal base and triangular faces. The triangular faces have the same size and shape and they connect the sides of the base to a common point called the apex.

Common Misconception
Students may forget to put radius into the formula when given diameter. Remind students of this when solving to find volumes of spheres.

4. A pyramid is shown, whose base is a rectangle with dimensions of 10 centimeters (cm) by 14 cm. The height of the pyramid is 17 cm. Determine the volume of the pyramid.

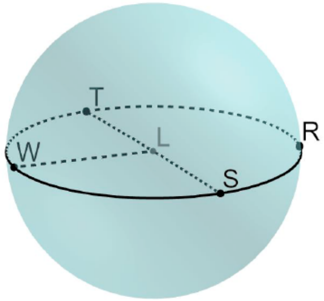


$$V = \frac{10 \cdot 14 \cdot 17}{3}$$
$$= \frac{2{,}380}{3}\ cm^3$$

9.6.2 Guided Instruction: Finding Volume (continued)

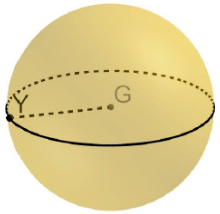
A ***sphere*** is a three-dimensional figure, where all points are equidistant from a point called the center.

A sphere with center L is shown, where $\overline{LW}$ is a radius and $\overline{TS}$ is a diameter. The circle that passes through the points $TRSW$ is known as a great circle since it is the largest possible circle that can be made around the sphere.



The volume formula for a sphere is $V = \frac{4}{3}\pi r^3$.

5. Find the volume of the sphere G, where $GY = 3$ using $\pi \approx 3.14$.



$$V = \frac{4}{3}\pi(3^3)$$
$$= \frac{4 \cdot 3.14 \cdot 27}{3}$$
$$= \frac{339.12}{3}$$
$$= 113.04 \ units^3$$

Use the following questions to guide student thinking on question 6:

- *What common features do the cone and cylinder share? (Same base.)*
- *Would the area of the bases of the cylinder be equal to the area of the base of the cone? (Yes.)*
- *How do you know that the diameters, radii, and the bases all have the same area in the decomposed shapes of the composite figure?*

Common Misconception
Students may forget to find the volume of both individual shapes and then add them together. Have students find the volume of each shape individually and label.

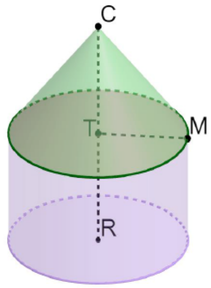
For a visual approach to these questions, have students make a table like the one shown below.

Three-Dimensional Figure	Work	Volume
Pyramid		
Rectangular Prism		
Composite Volume		

6. Composite figures are made up of more than one figure. The composite figure shown is a cone on top of a cylinder.

   Check all that must be true about the composite figure.

   - ☐ The heights have the same measure.
   - ■ *The diameters have the same measure.*
   - ■ *The radii have the same measure.*
   - ■ *The bases have the same area.*
   - ☐ The volume of this composite figure can be found using the formula $V = 2\pi r^2 h$.

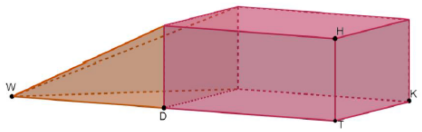


7. Discuss with your partner the statements that were false. Rewrite each false statement so that it would be true.

   The heights must be labeled or be able to be solved for.

   The volume of the composite figure can be found using the formula $V = \left(\pi r^2 \frac{h_{cone}}{3}\right) + (\pi r^2 h_{cylinder})$

8. A composite figure is shown, where $HT = 3.2$, $KT = 4.8$, $DT = 10.7$, and $WD = 11.1$. WD is the height of the pyramid. Find the volume of the composite figure.



$V = (l \cdot w \cdot h) + \left(\frac{l \cdot w \cdot h}{3}\right)$
$= (3.2 \cdot 4.8 \cdot 10.7) + \left(\frac{3.2 \cdot 11.1 \cdot 4.8}{3}\right)$
$= 164.352 + 56.832$
$= 221.184\ units^3$

9.6.3 Guided Practice: Finding Volume

Component Overview: *Students are moving to practicing without being given figures. Questions 1-2 require students to find the volume of a hemisphere. Have students be sure to find the volume of the sphere and then divide by* 2*.*

9.6.3 Guided Practice: Finding Volume

1. Find the volume of a hemisphere that has a diameter of 9.4 inches. Round your answer to the nearest thousandth.

$$V = \frac{2}{3}\pi r^3$$
$$= \frac{2}{3}\pi \cdot 4.7^3$$
$$= \frac{207.646\pi}{3}$$
$$= 217.446\ in^3$$

2. A composite figure consists of a hemisphere and a cone. The base of the cone and the great circle of the hemisphere have the same diameter. The radius of the hemisphere is $4\frac{3}{4}$ centimeters (cm). The height of the cone is $9\frac{1}{12}$ centimeters (cm). Find the volume of the composite figure using $\frac{22}{7}$ for π. Round your answer to the nearest thousandth.

$$V = \left(\frac{2}{3}\pi r^3\right) + \left(\pi r^2 \frac{h}{3}\right)$$
$$= \left(\frac{2}{3}\right)\left(\frac{19}{4}\cdot\frac{19}{4}\cdot\frac{19}{4}\right)\left(\frac{22}{7}\right) + \left(\frac{1}{3}\right)\left(\frac{109}{12}\right)\left(\frac{19}{4}\cdot\frac{19}{4}\right)\left(\frac{22}{7}\right)$$
$$= \left(\frac{2}{3}\right)\left(\frac{6859}{64}\right)\left(\frac{22}{7}\right) + \left(\frac{109}{36}\right)\left(\frac{361}{16}\right)\left(\frac{22}{7}\right)$$
$$= \frac{13{,}718}{192}\left(\frac{22}{7}\right) + \frac{39{,}349}{576}\left(\frac{22}{7}\right)$$
$$= \frac{75{,}449}{336} + \frac{432{,}839}{2016}$$
$$= \frac{885{,}533}{2016} = 439.252\ cm^3$$

Common Misconception
Students may substitute in the diameter instead of the radius into the formula. Students may also forget to divide the volume of a sphere by 2 *when finding the volume of a hemisphere.*

9.6.4 Your Turn!

Component Overview: *Students are asked to find the volumes of two figures without being given images of the figures.*

9.6.4 Your Turn!

1. Find the volume of a cylinder with a diameter of 18.3 millimeters (mm) and a height of 6.4 mm. Round your answer to the nearest thousandth.

   $V = \pi r^2 h$

   $= \pi \cdot 9.15^2 \cdot 6.4$

   $= 535.824\pi$

   $= 1{,}683.341\ mm^3$

2. Find the volume of a pyramid whose height is 15.7 inches (in.), and whose base is a rectangle with dimensions of 7.6 in. by 12.4 in. Round your answer to the nearest thousandth.

   $V = \frac{lwh}{3}$

   $= \frac{7.6 \cdot 12.4 \cdot 15.7}{3}$

   $= \frac{1479.568}{3}$

   $= 493.189\ in^3$

Use the following questions to guide students through a common misconception:

- *What is the difference between the radius and diameter?*
- *How is radius found when given the diameter?*
- *How can you use the volume of a cylinder formula to find the volume now that the radius is given?*

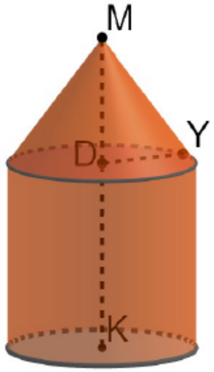
9.6.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students complete a question to measure mastery of the benchmark and the lesson target. Students should be allowed to access a calculator to complete 9.6.5 Wrap-Up.*

9.6.5 Wrap-Up: Ticket Out the Door

1. Find the volume of the composite figure if $MD = 17$, $DY = 8.5$, and $KD = 26$. Round your answer to the nearest hundredth.

$$V = (\pi r^2 h) + \left(\pi r^2 \frac{h}{3}\right)$$
$$= (\pi 8.5^2 \cdot 26) + \left(\pi 8.5^2 \frac{17}{3}\right)$$
$$= (\pi 72.25 \cdot 26) + \left(\frac{\pi \cdot 72.25 \cdot 17}{3}\right)$$
$$= (5{,}901.482) + (1{,}286.220)$$
$$= 7{,}187.70\ units^3$$



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

9.7 Volume in the Real World

Lesson Introduction			
Benchmark	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.	Learning Target	• I can use volume to solve problems with real-world applications.
Benchmark Coherence	Building On	Working Toward	
	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volumes of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. MA.7.GR.2.3 Solve mathematical and real-world problems involving volumes of right circular cylinders.	N/A	
Essential Questions	• How is volume used in the real-world? • How can I use the appropriate formula to find the volume of a three-dimensional figure? • How can I use volume to solve problems with real-world applications?		
Lesson Narrative	Lesson 7 adds to Lesson 6 by specifically having students work with real-world scenarios. By the start of this lesson, students should know the basic volume formulas and know how to substitute into a volume formula. Students work through a variety of real-world concepts to find volume or other needed information.		

Component Summary				
	9.7.1 Warm-Up (~5 min.)	Finding and explaining the error in an advertisement *(SE p. 89)*	9.7.3 Guided Practice (5-10 min.)	Solving problems involving volume with guidance *(SE p. 90)*
	9.7.2 Collaboration (10-15 min.)	Determining which packaging is best for a basketball *(SE p. 89)*	9.7.4 Your Turn! (10-15 min.)	Solving problems involving volume independently *(SE p. 91)*
	9.7.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 93)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	• (Optional) Basketball, Cardboard Boxes, Coffee Mug, Toy Truck, Tennis Ball Canister	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may incorrectly substitute values into formulas or use incorrect formulas.
On 9.7.4 Your Turn!, students may incorrectly find the volume of one tennis ball and/or the volume of the air.
Students may forget to find the area of the base when finding the volume of a pyramid.</td><td>On question 1 in 9.7.4 Your Turn!, consider having a canister of tennis balls to model the question.
Consider allowing students to work in partners on 9.7.4 Your Turn! if time runs short.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
Have students take turns working various steps of the questions. Students justify their work and settle disagreements if those arise. Encourage students to paraphrase what their partner says throughout the lesson.</td><td>MA.K12.MTR.7.1: Real-World Contexts
This lesson provides students with multiple opportunities to engage with real-world concepts by connecting math with real-world experiences and to use models to solve questions. The questions provided in this lesson give students a variety of real-world scenarios and require various levels of thinking.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider having models or manipulatives for this lesson. Find real-world objects that model a rectangular prism, cone, pyramid, sphere, and cylinder to have available for students to look at and use in class if necessary. Some examples are listed below.
Rectangular prism: any box
Cone: ice cream cones
Pyramid: paperweight, print a net and assembe
Sphere: basketball, baseball
Cylinder: paper towel roll
Providing tangible objects for students to reference can help students better visualize volume and can also help students see volume in the real-world.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

9.7.1 Warm-Up: Math Fail

Component Overview: *Students examine a picture and determine what math mistake is in the picture and how to fix the issue.*

9.7.1 Warm-Up: Math Fail

1. The clearance sign shown is what is known as a "math fail."

   a. What do you notice about the sign that makes this a math fail?

   Answers will vary.
   What is being asked to pay is more than the original price.

   b. Write a letter to the company explaining why this is a math fail and how to fix the issue.
   Answers will vary.
   This is a math fail because the total cost after the discount should not be more than the original price. Whoever wrote the discount tag should have taken 20% off of the original price of $\$6.99$ to get the new price to pay of $\$5.60$.

Consider having students quickly share their responses with classmates. Allow students to share anywhere they may have seen math errors like this in the real-world before. Discuss the financial impact this error could have on a company.

9.7.2 Collaboration: Packing Peanuts

Component Overview: *9.7.2 Collaboration provides students with an opportunity to engage with a real-world scenario to figure out which type of packaging is best for a basketball with a given diameter. Students find the volume of a rectangular prism, cylinder, and cube to answer the question by determining which packaging is best.*

9.7.2 Collaboration: Packing Peanuts

1. Josiah works for a sporting goods company in the shipping department. Josiah is trying to determine the best solution for packing a basketball with a diameter of 28.5 inches.

   a. What is the volume of the basketball?

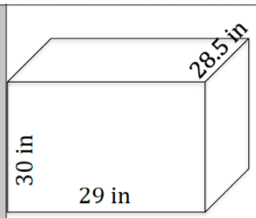
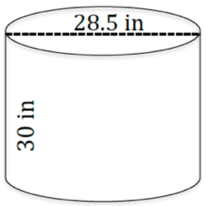
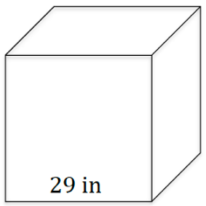
   $V = \frac{4}{3}\pi(14.25)^3$

   $V = 12{,}120.854\ inches^3$

   b. What size box do you think would be the best fit?

   Answer will vary.

   c. Find the volume of the three options of boxes below, from which Josiah can choose.

Rectangular Prism		Volume = <u>24,795 inches^3</u>
Cylinder		Volume = 19,138.190 inches^3
Cube		Volume = <u>24,389 inches^3</u>

Have students make a prediction as to which type of packaging would be the best fit and share with classmates to hear a variety of predictions.

Consider using a jigsaw strategy by assigning students either the rectangular prism, cylinder, or cube, and have students find the volume of their assigned figure. Bring the class together and have all groups share their volumes and have students record the results.

9.7.2 Collaboration: Packing Peanuts (continued)

d. Discuss with your partner how to determine the amount of space left over in the package after the basketball has been placed in it. Summarize your discussion.

Answers will vary.
Get the volume of the outer box being used and the volume of the basketball and subtract from each other to find the amount of space left over inside the package.

Have students share discussions with other groups or with the class before beginning question 1e.

e. After the basketball has been placed in the package, the remaining space is filled with packing peanuts. If each peanut is $1\ \mathrm{cm}^3$, which box will use the least amount of packing peanuts?

The difference between the volume of the cylindrical box and the volume of the basketball is the smallest, so the cylindrical box will need the least number of peanuts.

9.7.3 Guided Practice: Volume

Component Overview: *Students complete two practice questions on finding the volume of a rectangular prism and two cylinders.*

9.7.3 Guided Practice: Volume

1. Keon is building a raised garden bed that will have a width of 3.5 feet (ft), a depth of 2 ft, and length of 3 ft. If Keon fills the garden bed with soil, how much soil does he need to purchase?

   $V = l \cdot w \cdot h$

   $V = (3.5)(2)(3)$

   $V = 21 ft.^3$

   Keon needs to purchase $21\ ft^3$ *of soil.*

On question 2, the handles are not part of the diagram. Remind students to not include the handle when finding the volume.

2. Jasper wants to buy a new coffee mug, and he has narrowed it down to two options. The dimensions are given in inches (in.)

Option A	Option B
 Diameter: 2 in	 Radius: 1.25 in

In the event of a time crunch, assign half of the class Option A and assign the other half Option B.

   a. What is the volume, rounded to the thousandths place, of option A?

   $V = \pi(1^2)(5.25)$

   $V = 16.493\ in^3$

   b. What is the volume, rounded to the thousandths place, of option B?

   $V = \pi(01.25^2)(5)$

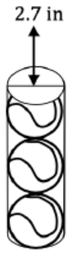
   $V = 24.544\ in^3$

9.7.4: Your Turn!

Component Overview: *Students work through a variety of application problems relating to finding the volume of a three-dimensional figure. On question 1, students find the volume of a canister and the air that is left over. In question 2, students compare the volumes of a sphere and a cone and must answer a statement. Students find the volume of a truck in question 3 and question 4 asks students to find the volume of a pyramid.*

9.7.4 Your Turn!

1. A tennis ball canister with three tennis balls is shown. The diameter of the canister is 2.7 inches (in) and the height is 8.1 in.



   a. What is the volume of the canister?

   $V = \pi(1.35^2)(8.1)$

   $V = 46.377\ in^3$

   b. What is the volume of 1 tennis ball?

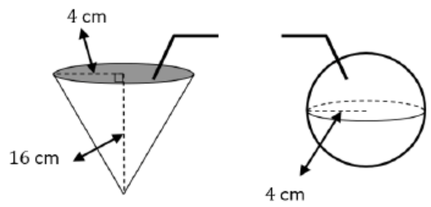
   $V = \frac{4}{3}\pi(1.35)^3$

   $= 10.306\ in^3$

   c. How much air is left in the canister after the tennis balls have been placed inside?

   $V = total\ volume - (volume\ of\ 3\ tennis\ balls)$
   $= 46.377 - 3(10.306)$
   $= 15.459\ in^3$

2. Club Cool in Epcot offers free samples of Coke products that are distributed worldwide. Flavors include Fanta Pineapple, Fanta Melon, Inca Cola, VegitaBeta, and Sparberry. You can choose to have your sample served in either of the containers shown.



Consider allowing students to continue working with partners and/or groups on this component of the lesson.

Common Misconception
Students may think the answer to question 1c is the answer to question 1b.

9.7.4: Your Turn! (continued)

Gian, Yanessa, and Jack made different conclusions about the drink containers. Who is correct and why?

Gian: The volumes are the same because the height of the cone is 4 times its radius, and both containers have the same radius.

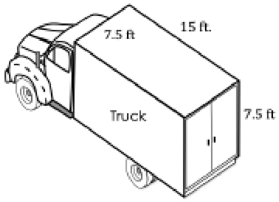
Yanessa: The volume of the sphere is greater than the cone's because $\frac{4}{3}\pi r^3$ is greater than $\frac{1}{3}\pi r^2 h$.

Jack: The volume of the cone is greater than the sphere's because the height is 4 times the radius of both containers.

Gian is correct because the height of the cone is 4 times its radius, and both containers have the same radius.

Tell students that the figures coming out of the cone and sphere are representing drinking straws and are not needed to solve the question.

3. Julian owns a moving company and has a truck that measures $15 \text{ ft} \times 7.5 \text{ ft} \times 7.5 \text{ ft}$.



He charges individuals by the number of boxes in the moving truck. If each box is a cube with side lengths of 1.5 ft, how many boxes can he fit in the truck?

$V = l \cdot w \cdot h$

$V = (7.5)(15)(7.5)$

$V = 843.75 ft.^3$

Volume of the truck$=843.75 ft.^3$

$V = a^3$

$V = 1.5^3$

$V = 3.375 ft.^3$

Volume of the box$=3.375 ft.^3$

$\frac{843.75}{3.375} = 250$ boxes

9.7.4: Your Turn! (continued)

4. The Pyramid of Cestius, pictured below, is located in Rome, Italy. The base of the pyramid is a square with side lengths of 30 meters and stands at a height of 36.5 meters. What is the volume of the pyramid?

$V = \frac{1}{3}Bh$

$V = \frac{1}{3}(30)^2(36.5)$

$V = 10{,}950m^3$

Common Misconception
Students may want to divide 843.75 by 1.5 to find the number of cubes. Monitor students' work to see if this occurs. If so, have students explain their thinking.

Consider asking the following for question 3.
- *How can you find the volume of a box?*
- *How can you find the total number of boxes?*

Remind students that when finding the volume of a pyramid, the B is the area of the base. Students may forget to square the side length of the base (30 meters) and not find the area of the square base.

9.7.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

9.7.5 Wrap-Up: Cool Down

1. Provide a real-world context that would require you to find the volume of a three-dimensional figure.
   Answers will vary.

9.8 Surface Area of Three-Dimensional Figures

Lesson Introduction

Benchmark Focus	MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.	Learning Targets	• I can solve mathematical problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres. • I can solve mathematical problems involving the lateral area of cylinders, pyramids, prisms, cones, and spheres.

	Building On	Working Toward
Benchmark Coherence	MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.	N/A

Essential Questions	• How can I solve mathematical problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres? • How can I solve mathematical problems involving the lateral area of cylinders, pyramids, prisms, cones, and spheres? • How can I derive the formulas for lateral area and surface area of a cylinder, pyramid, prism, cone, and sphere?
Lesson Narrative	This lesson guides students through the process of deriving formulas for lateral area and surface area of cylinders, prisms, and pyramids. The cone and sphere formulas are given. Students then work through a series of mathematical problems relating to finding the lateral areas and surface areas of various three-dimensional shapes. This lesson does not address real-world problems, as these will be addressed in Lesson 9.9.

Component Summary				
	9.8.1 Warm-Up (~5 min.)	Choosing which image doesn't belong *(SE p. 94)*	9.8.5 Guided Instruction (5-10 min.)	Finding lateral areas and surface areas of figures *(SE p. 97)*
	9.8.2 Collaboration (5 min.)	Determining lateral area and surface area *(SE p. 94)*	9.8.6 Guided Practice (5 min.)	Finding lateral area and surface area of a pyramid, prism, and sphere *(SE p. 98)*
	9.8.3 Activity (10 min.)	Exploring nets to determine lateral area and surface area *(SE p. 95)*	9.8.7 Your Turn! (5 min.)	Finding lateral area and surface area of a cone and cylinder *(SE p. 99)*
	9.8.4 Bring It Together (5 min.)	Formalizing the formulas for lateral area and surface area *(SE p. 96)*	9.8.8 Wrap-Up (~5 min)	Analyzing work finding surface area of a pyramid *(SE p. 99)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Lateral area • Surface area Additional Terms: N/A	• Nets for Gallery Walk • (Optional) Colored Pencils, Markers, Pens • (Optional) Various 3D Models	• This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (i.e. Word Wall, math cards, etc.).

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may use the incorrect formulas when finding surface area. • Students may use the surface area formula to find lateral area instead of the part of the formula that uses lateral area. • Students may not remember to divide the diameter by two when finding radius. • Students may calculate perimeter instead of area. • Students may not round final answers to the correct place value.	• Consider paring down parts of the 9.8.5 Guided Instruction, 9.8.6 Guided Practice, or 9.8.7 Your Turn! in the event of a time crunch by allowing student choice of individual questions. • Consider having three-dimensional models and/or nets as another visual for students.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Which One Doesn't Belong? 9.8.1 Warm-Up in this lesson is a Which One Doesn't Belong. Allow students to share their specific reasonings as to which one doesn't belong in the set. Pay specific attention to students' vocabulary here and create an environment that is no risk and that is accessible to all students.	**MA.K12.MTR.4.1: Discussion and Reflection** This lesson has multiple opportunities for student discussion built in and structured. 9.8.2 Collaboration requires that students discuss to develop the formulas for surface area, and the 9.8.4 Bring It Together has students discussing the various formulas. MA.K12.MTR.4.1 also has students discuss and compare the efficiency of methods, which is also present within the lesson.
Differentiate Instruction	9.8.2 Collaboration and 9.8.3 Activity are appropriate to differentiate instruction for students. As students derive formulas, some may need more scaffolding. Provide digital visuals using technology. Consider using GeoGebra to provide the visuals for students.	

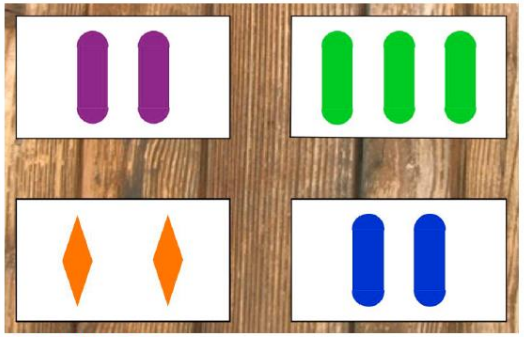
Lesson Notes:

9.8.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students begin a low-floor, high-ceiling task by stating which card does not belong in the set and why.*

9.8.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong? Explain your answer.



Answers will vary.

The image with the 2 orange rhombi does not belong because the rhombi are not composite figures.

The image with the purple figures does not belong because purple is not a primary color.

The image with the green figures contains 3 composite figures.

The image with the 2 blue figures because in the second row, they are not rhombi.

Allow students to think about which card in the set does not belong. Provide an opportunity for several students to share their responses.

For a kinesthetic approach to 9.8.1 Warm-Up, have students stand in separate corners of the room and hear a response or two from each group. Students will stand in the corner that matches their response.

A - top left (purple card)
B - top right (green card)
C - bottom left (orange card)
D - bottom right (blue card)

9.8.2 Collaboration: Using Nets to Find Surface Area

Component Overview: *Students will begin working with a partner to determine formulas for lateral area and surface area of various three-dimensional figures. Assign a three-dimensional figure to each group and give time for students to derive the formula using the prompts in the questions.*

9.8.2 Collaboration: Using Nets to Find Surface Area

The **lateral area** of a three-dimensional figure is the sum of the area of the figure's faces.

The **surface area** of a three-dimensional figure is the sum of the lateral area of the figure's faces and the area of the figure's bases.

1. Work with your partner to determine a formula for the lateral area and the surface area of the figure you have been given. Complete the table.

Our figure was:				
Dimension				
Variable Used				

Answers will vary. See the table in the Gallery Walk below.

Students will not be given specific numbers to use, so only variables will be used here.

Teachers may need to review the difference between the faces and bases of a figure.

Teachers assign each pair one of the six figures from the 9.8.3 Activity to determine the surface area. Each pair can be given a larger version of the figures or use the figures in the table. Once each pair has completed 9.8.2 Collaboration, then place students in groups based on their figure to create information for the Gallery Walk.

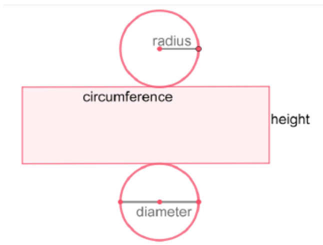
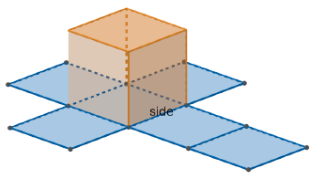
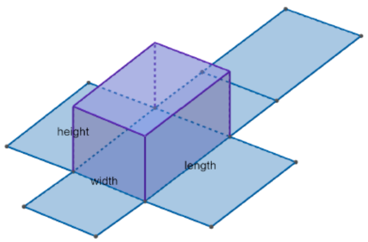
Teachers may need to explain why the student must define the variables they used.

9.8.3 Activity: Gallery Walk

__Component Overview:__ Students are recording the formulas from the other groups onto their individual papers. Post the students' work around the classroom and have students complete a Gallery Walk. Give students no more than one to two minutes per station. Create a structure that allows students to move through this as efficiently as possible.

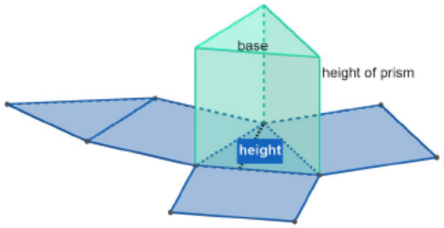
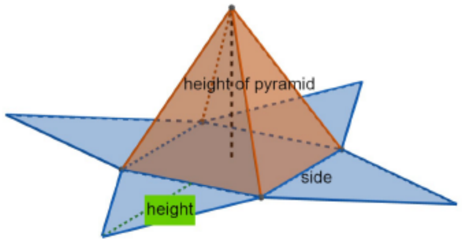
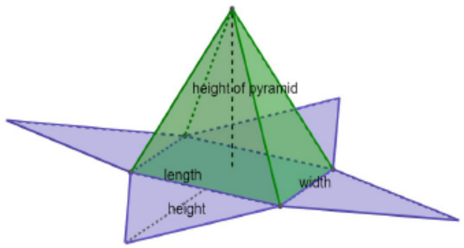
9.8.3 Activity: Gallery Walk

1. During the gallery walk, record the formulas. Make any useful notes on each figure's net.

Figure	Net
Cylinder Lateral Area = $2\pi rh$, where r = radius and h = height. Surface Area = $2(\pi r^2 + \pi rh)$	
Cube Lateral Area = $4s^2$, where s = side. Surface Area = $6s^2$	
Rectangular Prism Lateral Area = $2(lh + wh)$, where l = length, h = height and w = width. Surface Area = $2(lh + wh + lw)$	

Students may find during the Gallery Walk that different formulas were created. This will provide a chance for a rich conversation for the teacher to have with the class. Formulas are meant to make finding something easier. Students should not be forced to memorize a formula.

9.8.3 Activity: Gallery Walk (continued)

Triangular Prism Lateral Area = $3bh_p$, where b = base and h_p = height of prism. Surface Area = $3bh_p + bh$, where h = height of triangular base.	
Square Pyramid Lateral Area = $2sh_s$, where s = side. Surface Area = $2sh_s + s^2$	 The height of the triangle that is a face of a pyramid is also known as slant height, h_s.
Rectangular Pyramid Lateral Area = $h_s(l+w)$, where l = length and w = width. Surface Area = $h_s(l+w) + lw$	 The height of the triangle that is a face of a pyramid is also known as slant height, h_s.

How does the formula for the surface area triangular prism differ from the formula for the surface area of a rectangular prism?

How does the formula for the square pyramid prism differ from the formula for the rectangular pyramid?

9.8.4 Bring It Together: Formulas

Component Overview: *Students are discussing the similarities and differences between the formulas that were derived in 9.8.3 Activity when compared to the formulas given in the chart. Students are also introduced to the formulas for surface area of a sphere and a cone for the first time in this lesson.*

9.8.4 Bring It Together: Formulas

Using the net of three-dimensional objects can help to find the lateral area and surface area. The net of a cone and a sphere make deriving the formulas for surface area from the net more challenging. A chart of formulas is given, where SA represents surface area.

Figure	Surface Area	Variables
Prism	$SA = 2B + Ph$	B = area of base, P = perimeter, h = height
Cylinder	$SA = 2B + Ch$ Or $SA = 2B + Ph$	B = area of base, C = circumference, P = perimeter, h = height)
Cone	$SA = B + \pi r h_s$	B = area of base, r = radius, h_s = slant height
Regular Pyramid (a pyramid whose base is a regular polygon)	$SA = B + \frac{1}{2} P h_s$	B = area of base, P = perimeter, h_s = slant height
Sphere	$SA = 4\pi r^2$	r = radius

This component provides an opportunity for students to discuss how the formulas differ. Provide students time to discuss the similarities and differences in the formulas.

Why does a sphere not have a lateral area circled? (There is no specific lateral area for a sphere.)

1. The formulas in the table may not match the formulas recorded during the gallery walk. Which figures have formulas that are different?
   Answers will vary. All the formulas are different because the formulas in the gallery walk were written in terms of the actual area of each figure's base versus B, the letter used to represent area of the base of any figure. In addition, the formulas in the gallery walk were not written in terms of P, the perimeter of the base.

9.8.4 Bring It Together: Formulas (continued)

2. There are many ways to write and to understand formulas. Discuss with your partner which formulas make more sense to you. Discuss how the formulas in the formula chart can be used to help you remember how to find surface area. Summarize your discussion.
Answers will vary. The surface area of a prism, $SA = 2B + Ph$ *makes more sense because it appears to be more direct and simpler to work with.*

3. The chart does not have a formula for lateral area. For each of the figures, except for spheres, circle the part of the surface area formula that gives the lateral area.
See the table above.

How can you tell which part of the formula is the lateral area? (The part that does not relate to the base of the figure.)

Do not spend too much time on having students circle the lateral area, as this will take away from the discussion.

9.8.5 Guided Instruction: Lateral Area and Surface Area

Component Overview: *Students work through questions finding the lateral area and the surface area with teacher support. This component provides a mixture of questions for students to work through with support from the teacher.*

9.8.5 Guided Instruction: Lateral Area and Surface Area

1. A square pyramid has a slant height of 7 inches, and the area of the base is 16 square inches. What is the surface area, in square inches?

   $SA = B + \frac{1}{2}Ph_s$

   $SA = 16 + \frac{1}{2}(16)(7)$

   $SA = 16 + (8)(7)$

   $SA = 16 + 56$

   $SA = 72\ in^2$

2. The surface area of a sphere is 144π feet. What is the radius of the sphere?

   $SA = 4\pi r^2$
   $4\pi r^2 = 144\pi$
   $r^2 = 36$
   $r = 6\ ft$

3. If a cylinder has a height of 16 centimeters and a radius of 8 centimeters.
   a. Find the lateral area. Leave your answer in π form.
      $LA = Ch$
      $LA = 2\pi(8)(16)$
      $LA = 256\pi\ cm^2$

   b. Find the surface area. Leave your answer in π form.
      $SA = 2B + Ch$
      $SA = 2(\pi 8^2) + 2\pi(8)(16)$
      $SA = 128\pi + 256\pi$
      $SA = 384\pi\ cm^2$

Common Misconception
Students may mistakenly find the area instead of the perimeter. When necessary, review how to calculate the perimeter of a figure.

How does question 2 differ from question 1? (Surface area is given, and students must find the radius.)

What does it mean to leave your answers in terms of π? (Do not multiply by the decimal representation of π when simplifying the formula.)

On question 3b, have students streamline the process by substituting the lateral area into the surface area formula.

9.8.5 Guided Instruction: Lateral Area and Surface Area (continued)

4. A right triangular prism has side lengths of $8, 15$, and 17 feet (ft), where 17 ft is the hypotenuse of the triangle. The prism has a height of 7 ft. What is the surface area, in square feet, of the triangular prism?

$SA = 2B + Ph$

$SA = 2\left(\frac{1}{2} \cdot 8 \cdot 15\right) + (40)(7)$

$SA = 120 + 280$

$SA = 400\, ft^2$

On question 4, encourage students to draw and label an image of a triangular prism prior to finding the surface area.

9.8.6 Guided Practice: Lateral Area and Surface Area

Component Overview: *Students are working through other mathematical questions relating to finding lateral and surface area.*

9.8.6 Guided Practice: Lateral Area and Surface Area

1. A pyramid has a square base whose area is 49 square feet with a height of 11.5 feet. Find the lateral area, in square feet, of the pyramid.

   $LA = \frac{1}{2}Ph_s$
   $LA = \frac{1}{2}(28)(\sqrt{144.5})$
   $LA = 168.291\ ft^2$

2. A rectangular prism has a length of 12.5 inches (in), a width of 8 in., and a height of 5 in. Determine the surface area, in square inches, of the rectangular prism.

   $SA = 2B + Ph$
   $SA = 2(8)(12.5) + (41)(5)$
   $SA = 405\ in^2$

3. The radius of a sphere is 18.7 feet. Find, in square feet, the surface area of the sphere. Round your answer to thousandths place.

   $SA = 4\pi r^2$
   $SA = 4\pi(18.7)^2$
   $SA = 4394.334\ ft^2$

To provide students a choice, allow students to use either the formulas that were derived in 9.8.3 Activity or have students use the formulas that were given in the chart.

Students are required to find perimeter if using the formulas from the chart. Students must remember that finding the perimeter requires adding all the sides of the base.

9.8.7 Your Turn!

Component Overview: *Students work independently through two questions related to finding the lateral area of a cone and the surface area of a cylinder.*

9.8.7 Your Turn!

1. Find the lateral area, in square meters, of a cone that has a radius of 3 meters (m) and a slant height of 8.5 m. Round your answer to thousandths place.
   $LA = \pi r h_s$
   $LA = \pi(3)(8.5)$
   $LA = 80.111\ m^2$

2. A cylinder has a radius of 5 inches (in) and a height of 3.75 in. Find the surface area, in inches, of the cylinder. Round your answer to thousandths place.
   $SA = 2B + Ch$
   $SA = 2\pi(5)^2 + 2\pi(5)(3.75)$
   $SA = 50\pi + 37.5\pi$
   $SA = 274.889\ in^2$

How is finding lateral area different from finding surface area? (Lateral area does not involve the bases.)

Common Misconception
Students may not round to the correct place value and may require additional help and scaffolding with rounding. Provide this to students as necessary. This may be an issue with students' final answers.

9.8.8 Wrap-Up: Error Analysis

Component Overview: *Students complete an error analysis of a question involving the surface area of a square pyramid.*

9.8.8 Wrap-Up: Error Analysis

1. Cody and Mila are finding the surface area of a square pyramid whose base area is 144 square inches and slant height is 16 inches. Their work is shown in the table.

Cody's Work	Mila's Work
$SA = B + \frac{1}{2}Ph_s$	$SA = B + \frac{1}{2}Ph_s$
$SA = 144 + \frac{1}{2}(48)(16)$	$SA = (144) + \frac{1}{2}(144)(16)$
$SA = 528$ square inches	$SA = 1296$ square inches

a. Whose work is correct?
Cody is correct.

b. Write a note to the student whose work is incorrect describing the mistake the student made.
Mila, your process is correct; you used the correct formula. However, in your first step, you substituted, 144, the measure of the area of the base of the pyramid instead of, 48, the perimeter of the base.

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

For an extension, change a part of Cody's work that would make his work incorrect.

9.9 Surface Area in the Real World

Lesson Introduction

Benchmark	MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.	Learning Targets	• I can solve real-world problems involving the surface areas of cylinders, pyramids, prisms, cones, and spheres. • I can solve real-world problems involving the surface areas of composite figures.

Benchmark Coherence	Building On	Working Toward
	MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.	N/A

Essential Questions	• How can I solve real-world problems involving the surface areas of cylinders, pyramids, prisms, cones, and spheres? • How can I solve real-world problems involving the surface areas of composite figures? • How is surface area used in the real-world?
Lesson Narrative	Lesson 9 builds on Lesson 8 in working with the surface areas of various three-dimensional figures. The purpose of Lesson 9.9 is to specifically work with real-world problems and scenarios and to be able to apply the surface area formulas in a real-world context. This lesson will guide learning in Lesson 11 on dimension changes.

Component Summary				
	9.9.1 Warm-Up (~5 min.)	Analyzing a math magic trick *(SE p. 100)*	9.9.3 Guided Instruction (10-15 min.)	Solving problems on frosting cakes using surface area *(SE p. 102)*
	9.9.2 Collaboration (10-15 min.)	Solving a packaging problem using surface area *(SE p. 101)*	9.9.4 Guided Practice (5-10 min.)	Using surface area to solve problems *(SE p. 103)*
	9.9.5 Wrap-Up (~5 min.)	Ticket Out the Door on real-world surface area *(SE p. 104)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	• (Optional) Colored Pens/Pencils	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>On 9.9.2 Collaboration question 3, students may forget to divide by 100 to find the cost of the packaging.
On 9.9.3 Guided Instruction question 2, students may try to find the surface area of the rectangular prism instead of lateral area.
Students may multiply by $\frac{1}{3}$ instead of dividing on 9.9.4 Guided Practice, question 1.</td><td>Consider letting students work with either partners or groups on the whole lesson, with the exceptions of the 9.9.1 Warm-Up and 9.9.5 Wrap-Up, as these questions are more complex than the previous lesson's questions.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Number Talk
Consider using the 9.9.1 Warm-Up as an opportunity to discuss the process of how to find the "magic number." Have students rely on the structure and the algorithm and discuss why the ending result is always 5.</td><td>MA.K12.MTR.6.1: Reasonableness
Many of the problems in this lesson are real-world scenarios and require students to take out parts of the formulas to find the correct answers. Have students discuss and work to determine whether a solution is reasonable for a given context.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider making the 9.9.2 Collaboration, 9.9.3 Guided Instruction, and the 9.9.4 Guided Practice three separate groups. Assign each student to one of the three sections. Give students a set amount of time to complete the component. Then, create new groups of three made up of one student from each of the three sections above (9.9.2 Collaboration, 9.9.3 Guided Instruction, and 9.9.4 Guided Practice). Students share their results with their group and become "experts" on their component.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

9.9.1 Warm-Up: Math Magic

Component Overview: *Students complete a "magic math trick" and then try to justify the "trick" by creating an equation.*

9.9.1 Warm-Up: Math Magic

Arthur Benjamin (b. 1961) is a professor of mathematics at Harvey Mudd College in California. Dr. Benjamin is often referred to as a "mathemagician" because he is known for his quick mental computational skills. Dr. Benjamin's TED talks have been viewed millions of times on the internet.

If time permits, have students try the steps using a number greater than 10 to see if the result is still 5.

Try one of Dr. Benjamin's math "magic tricks."

- Pick a number between 1 and 10.
- Double your number.
- Add 10 to your number.
- Divide by 2.
- Subtract the number you started with originally.

1. What number did you get?
   5

2. Compare your number with your partner's number. What do you notice?
   The answers are the same.

3. Why do you think the "trick" works?
   Answers will vary. Sample response: Regardless of the first number, the process is the same.

Pick a number between 1 and 10.	x
Double your number.	$2x$
Add 10 to your number.	$2x + 10$
Divide by 2.	$x + 5$
Subtract the number you started with originally.	5

9.9.2 Collaboration: The Best Package

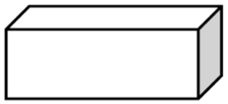
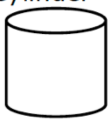
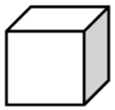
Component Overview: *Students work through a scenario where three types of packaging are being used. Students will determine the surface areas of the types of packaging and then determine which type of packaging will be the cheapest. Allow students to work in partners or groups to complete this component.*

9.9.2 Collaboration: The Best Package

Have students make a prediction to see which type of packaging they think will be the largest and smallest.

Manuel is a graphic designer and works with various types of packaging. For the artwork, Manuel needs a container with a surface area of at least 250 square inches (in.) to ship the company's products.

1. The three types of packaging that the company has in stock are shown in the table. Find the surface area for each type of packaging.

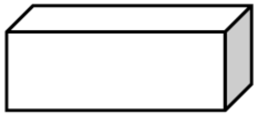
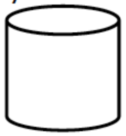
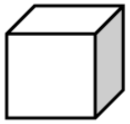
Packaging	Dimensions	Surface Area
Rectangular Prism 	l: 7 in. w: 10 in. h: 4 in.	$SA = 2B + Ph$ $SA = 276\ in^2$
Cylinder 	d: 8 in. h: 8 in.	$SA = 2B + Ch$ $SA = 301.593\ in^2$
Cube 	s: 12 in.	$SA = 6s^2$ $SA = 864\ in^2$

2. Rank the packaging's surface area from smallest to largest.

Rank	Packaging
1 (smallest)	*Rectangular Prism*
2	*Cylinder*
3 (largest)	*Cube*

9.9.2 Collaboration: The Best Package (continued)

3. The prices for packaging are listed in the table. Based on the surface area of each package, find the cost of each type of packaging.

Packaging	Pricing	Cost of Packaging
Rectangular Prism 	$1.75 per 100 square inches	$\frac{276}{100} = 2.76$ $1.75 \cdot 2.76 = 4.83$ $4.83 *per package*
Cylinder 	$2.25 per 100 square inches	$\frac{301.6}{100} = 3.016$ $2.25 \cdot 3.016 = 6.786$ $6.79 *per package*
Cube 	$1.25 per 100 square inches	$\frac{864}{100} = 8.64$ $1.25 \cdot 8.64 = 10.8$ $10.80 *per package*

4. If Manuel wants to use the cheapest packaging possible for his client, which packaging should he choose?
Manuel should choose the rectangular prism shaped package. It costs $4.83 *per package and the cheapest packaging.*

When students are completing question 3, monitor that students are dividing the surface area by 100 to start calculating the cost of the packaging.

9.9.3 Guided Instruction: Real-Life Applications of Surface Area

Component Overview: *9.9.3 Guided Instruction provides a scenario where students are determining how much frosting to use on two types of cakes. In questions 1-3 students compare the amount of frosting needed for cakes that are different figures and composite figures.*

9.9.3 Guided Instruction: Real-Life Applications of Surface Area

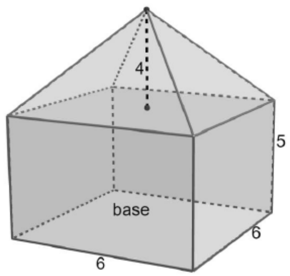
Jeni owns a bakery and is making two cakes.

Cake A	Cake B
Cake A is in the shape of a rectangular prism that is 3 inches thick, 9 inches long, and 12 inches wide.	The cake is in the shape of a cylinder and has a diameter of 8 in. The cake is composed of two layers, each 3 in. tall.

1. When Jeni frosts the cake, she applies the frosting to all surfaces except for the bottom of the cake. She also frosts between each layer. Determine the amount of frosting Jeni needs to apply to each cake. Show your work.

Cake A	Cake B
$SA = B + Ph$ $SA = (12 \cdot 9) + (42 \cdot 3)$ $SA = 108 + 126$ $SA = 234\ in^2$	$SA = 2(B + Ch)$ $SA = 2(\pi(4)^2 + 2\pi(4)(6))$ $SA = 2(40\pi)$ $SA = 251.327\ in^2$

2. Jeni makes a two-layer cake, as shown, where the measurements are in inches (in.) She will frost the whole cake except for the base of the bottom layer. She also adds frosting between layers.
   a. Determine the total surface area that Jeni needs to frost.



The cake is a composite figure of a pyramid and a rectangular prism.

SA of Pyramid + *LA of Rect. Prism* = *Total SA*

$SA = B + \frac{1}{2}Ph_s$
$h_s = 5$
$SA = 36 + \frac{1}{2}(24)(5)$
$SA = 96\ in^2$

$LA = Ph$
$LA = (24)(5)$
$LA = 120\ in^2$

Total $SA = 96 + 120 = 216\ in^2$

Consider assigning students either Cake A or Cake B. After students have worked their question, pair each student in Group A with a student from Group B to share their results with each other.

Before students work on question 1, ask the following to prompt student thinking and responses:

- *If the cakes are not frosted on the bottom, how does this change the formula to find the surface area?*

9.9.3 Guided Instruction: Real-Life Applications of Surface Area (continued)

b. Jeni's recipe for buttercream icing typically frosts a round cake with an 8 in. diameter that is 1.5 in. tall. How many batches of frosting will Jeni need to frost the two-layer cake in part A?

SA (of one batch) $= (B + Ch)$

$= \pi(4)^2 + 2\pi(4)(1.5)$

$= 87.965\ in^2$

Number of batches $= \frac{SA\ of\ cake}{SA\ of\ one\ batch} = \frac{216}{87.965} = 2.45$ batches

Jeni will need 3 batches of frosting to frost the two layer cake.

Common Misconception
Students may include the base of the prism when finding the surface area that will be iced with frosting. Monitor students' work and ask probing questions about whether to include the base.

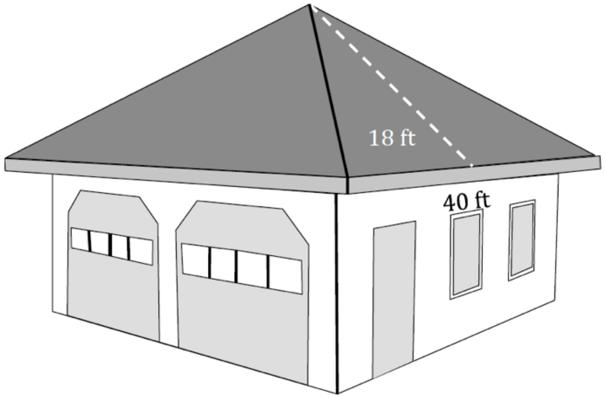
What part of the cake is frosted when Jeni frosts the first layer?

9.9.4 Guided Practice: Real-Life Applications of Surface Area

Component Overview: *Students work through two real-life application questions involving the roofs of a garage and an art gallery.*

9.9.4 Guided Practice: Real-Life Applications of Surface Area

1. Lourdes' garage is in the shape of a square pyramid. The garage is shown, with dimensions given in feet (ft).



Lourdes is replacing the roof and installing shingles. In roofing terminology, a "square" is 100 square feet. Composition shingles come in bundles that cover $\frac{1}{3}$ of a square.

How many bundles should Lourdes buy to cover the roof?

Excluding the area of the base of the pyramid, use the lateral area formula, $LA = \frac{1}{2}Ph_s$.

$LA = \frac{1}{2}(160)(18) = 1{,}440\ ft^2$

They will need $\frac{1{,}440}{100} = 14.4$ *"squares"*

They will need $\frac{14.4}{\left(\frac{1}{3}\right)} = 43.2$*, which is about* 44 *bundles of shingles.*

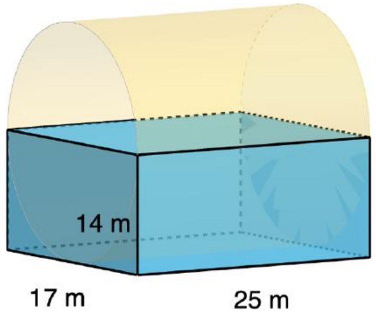
If students determine they need $4{,}320$ *bundles, then they failed to divide the lateral area by* 100 *to find the number of squares in the lateral area.*

If students determine they need 4.8 *or* 5 *bundles, then they multiplied by* $\frac{1}{3}$ *instead of dividing by* $\frac{1}{3}$ *(or multiplying by* 3*).*

Why is lateral area calculated instead of surface area to find the number of shingles? (The roofing is only for the part of the roof outside of the house, which is lateral area.)

9.9.4 Guided Practice: Real-Life Applications of Surface Area (continued)

2. An architect is designing an art gallery. She used computer software to create a rough sketch of her idea. Her sketch is shown with measurements in meters (m). She would like for the art gallery to be all glass. To determine the estimated cost, she would need to find the area of all of the surfaces except for the floor. Determine the total surface area, in square meters, of the glass in the art gallery. Round to the nearest thousandth.



This is a composite figure of $\frac{1}{2}$ of a cylinder and the lateral area of a rectangular prism. Assume that the height of the rectangular prism is 14 meters.

$\frac{1}{2}$ SA of Cylinder **+** *LA of Rect. Prism* **=** *Total SA*

$SA = \frac{1}{2}(2B + Ch)$
$SA = \pi r^2 + \pi rh$
$SA = \pi(8.5)^2 + \pi(8.5)(25)$
$SA \approx 284.75\pi\ m^2$

$LA = Ph$
$LA = (84)(14)$
$LA = 1{,}176\ m^2$

Total $SA =$ $2{,}070.569\ m^2$

As students are working through question 2, consider asking students the following question to prompt thinking:

- *Why does the surface area of the cylinder need to be multiplied by $\frac{1}{2}$?*

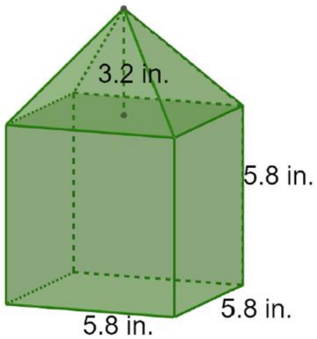
For students who finish early, challenge them to create a composite shape and scenario and then solve the question. If extra time allows, provide this question to other students who finish early.

9.9.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students complete a Ticket Out the Door to demonstrate mastery on this lesson's learning target. Students should be allowed to access a calculator to complete 9.9.5 Wrap-Up.*

9.9.5 Wrap-Up: Ticket Out the Door

1. Jillian was building a birdhouse. Some of the dimensions, in inches (in.), of the birdhouse are shown. Determine the total amount of wood, in square inches, that Jillian will need to build the outside of the birdhouse.



Students only need to find the lateral area of the pyramid and only use one base when finding the surface area of the rectangular prism.

LA of Pyramid **+** *SA of Rect. Prism with 1 base* **=** *Total SA*

$LA = \frac{1}{2}Ph_s$

$LA = \frac{1}{2}(23.2)(\sqrt{18.65})$

$LA = 50.095$

$SA = B + Ph$

$SA = 33.64 + 134.56$

$SA = 168.2$

$SA = 218.295\ in^2$

9.10 Determining How Dilations Affect Area

Lesson Introduction

Benchmark	MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.	Learning Target	• I can discover the relationship between area and the scale factor of a dilation.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• What is the relationship between area and the scale factor of a dilation? • How does the scale factor of a figure affect the area?
Lesson Narrative	The lesson gives students the opportunity to discover how the area of a preimage and an image relate to dilations with given scale factors. 9.10.2 Exploration guides students to make conjectures to discover the relationship between the area and the scale factor of the dilation. After 9.10.2 Exploration, students work through two components to gain practice to master this benchmark.

Component Summary	9.10.1 Warm-Up (~5 min.)	Speculating about a directional sign *(SE p. 105)*	9.10.3 Guided Practice (10-15 min.)	Solving problems using scale factor and area *(SE p. 109)*
	9.10.2 Exploration (10-15 min.)	Discovering the relationship between the scale factor of a dilation and area *(SE p. 105)*	9.10.4 Your Turn! (5-10 min.)	Finding the area of a dilated circle *(SE p. 110)*
	9.10.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 111)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• (Optional) Colored pencils/Pens • (Optional) Graphing Utility	• N/A

Teacher Tips	**Common Misconceptions** • Students may mistakenly think that the scale factor is the proportion that the area will increase or decrease by. • Students may incorrectly identify a scale factor of less than 1 as an enlargement. • On 9.10.4 Your Turn! question 1, students may incorrectly find the radius of the image.	**Things to Consider** • Consider allowing students to use a calculator to compute the coordinates that have fractions. • Consider showing applets in GeoGebra or another program that provide students with a visual approach.
	Literacy and Language Routines **Take Turns** Provide students an opportunity to work 9.10.2 Exploration with either a partner or a small group. The questions and prompts lend themselves for students to take turns working through the guided discovery. Create a structure where students explain their work and their conjectures.	**Mathematical Thinking and Reasoning (MTR) Standard** **MA.K12.MTR.1.1: Participate in Effortful Learning** MA.K12.MTR.1.1 states that students are expected to help and support one another when completing mathematical tasks and to analyze and make sense of the questions. 9.10.2 Exploration gives students the opportunity to practice and apply the skills of active participation.
Differentiate Instruction	Consider using embedded visuals into 9.10.2 Exploration to help guide students through the component. GeoGebra has a plethora of applets that model scale factor and area to help students better visualize how the scale factor of a dilation affects the area of a two-dimensional figure. Within GeoGebra, search “scale factor and area” and choose an applet that best fits student needs.	

Lesson Notes:

9.10.1 Warm-Up: Where Is Here?

Component Overview: *Students complete three questions involving a picture.*

9.10.1 Warm-Up: Where is Here?

A picture of a directional sign is shown.

1. What do you notice about the sign?
   Answer will vary.
   The sign says, "Here". It gives no true direction.

2. Where do you think the picture was taken?
   Answer will vary.
   In a city near a river, maybe a European city.

3. The Earth's average polar circumference is $24{,}859$ miles. What do you think the sign means?
   Answer will vary.
   It may mean that if someone walks the complete circumference of the Earth's surface, they will pass this sign.

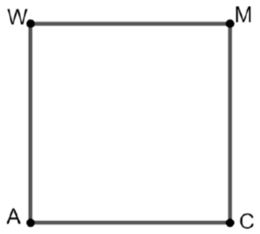
Consider also asking students, "What do you wonder about the sign?" Allow them the opportunity to share out their answers to gather group discussion and thoughts around this picture, since there is not one correct answer for these questions.

9.10.2 Exploration: Scale Factor and Area

Component overview: *9.10.2 Exploration begins with a one-unit square where students are required to find the area, perform a dilation to discover a pattern, and write a conjecture relating the area of a figure to the scale factor. Then students refine their conjectures by applying it to larger squares, rectangles, and parallelograms. Students are expected to apply previous knowledge of dilations.*

9.10.2 Exploration: Scale Factor and Area

1. Square $AWMC$ is shown, where $WM = 1$ unit.



a. Determine the area of $AWMC$.
$A = 1\ unit^2$

b. $AWMC$ is dilated using point A as the center of dilation using each of the given scale factors. Complete the table.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation	$\frac{1}{2}$ unit	$\frac{3}{4}$ unit	2 units	3 units	4 units
Area of Dilated Square	$\frac{1}{4}\ unit^2$	$\frac{9}{16}\ unit^2$	$4\ unit^2$	$9\ unit^2$	$16\ unit^2$

c. Write a conjecture about the relationship between the length of the scale factor and the value of the area.
The value of the area of $AWMC$ is equal to the square of the measure of the scale factor.

d. Determine the area of $AWMC$ if $WM = 4$ units.
$A = 16\ unit^2$

Students should be collaborating throughout the component. Monitor and listen for common student misconceptions to identify which students surface those misconceptions and which students may need help with clarifications.

Consider asking students the following questions as they are working on the table in question 1b:

- *Why is the side length of the dilation the same as the scale factor? (The figure is a square, and the preimage is 1.)*
- *How do you find the area of the dilated square?*

9.10.2 Exploration: Scale Factor and Area (continued)

2. Suppose square $DTSC$ is dilated using point D as the center of dilation for each of the given scale factors. Complete the table for $DTSC$ given that $TS = 4$ units.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation	$2\ units$	$3\ units$	$8\ units$	$12\ units$	$16\ units$
Area of Dilated Square	$4\ units^2$	$9\ units^2$	$64\ units^2$	$144\ units^2$	$256\ units^2$

a. Review your conjecture from the previous problem and make any revisions, if needed.
The value of the area of $DTSC$ is equal to $(side\ length \times scale\ factor)^2$.

b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

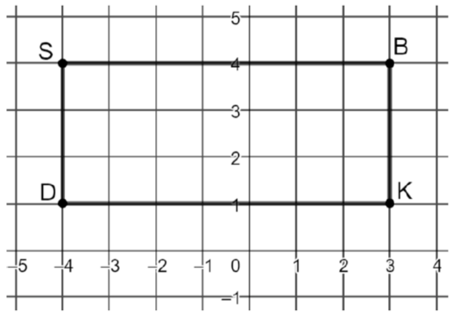
Summary of Partner's Conjecture	Partner's Initials
Answers will vary.	

c. Review your conjecture and make any revisions, if needed.
Answers will vary.

For a kinesthetic approach to 9.10.2 Exploration, consider dividing the classroom into three groups, one for the square, the rectangle, and the parallelogram. Have students become an expert on their figure. Regroup the classroom into groups of three where each student shares their results of their questions.

Ask the following questions to prompt student thinking:
- *Does the conjecture need to be revised? Why or why not?*

3. Rectangle $SBKD$ is shown.



a. Using the coordinate grid, find the area of the rectangle.

Area=_21_ square units

b. Dilate $SBKD$ by a scale factor of 3 centered at the origin. Write the new coordinates in the table.

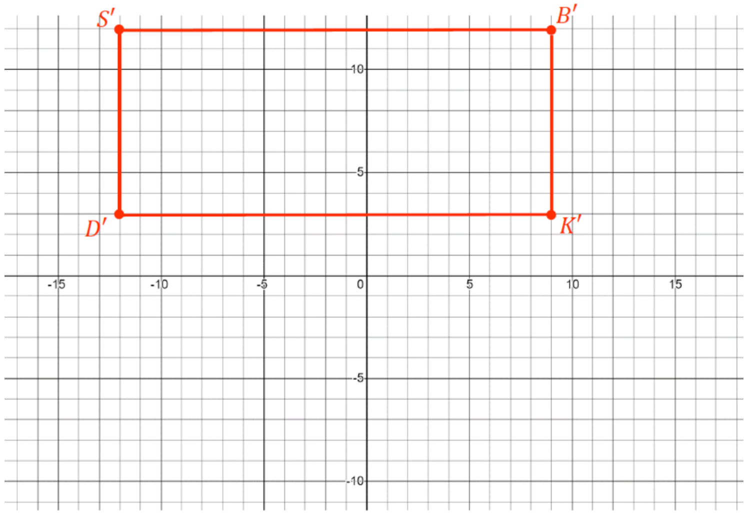
Point	Preimage	Image
S	$(-4,4)$	$(-12,12)$
B	$(3,4)$	$(9,12)$
K	$(3,1)$	$(9,3)$
D	$(-4,1)$	$(-12,3)$

Does the conjecture need to be revised? Why or why not?

What does the dilation do to the figure? How is this different than a translation?

9.10.2 Exploration: Scale Factor and Area (continued)

4. Graph $S'B'K'D'$ on the coordinate plane.



a. The area of rectangle $S'B'K'D'$ is __189__ square units.

b. Does the conjecture you wrote about the relationship between area and scale factor when dilating a square hold true for rectangle $SBKD$?

Answer will vary.

No, for a rectangle, the area of the dilated rectangle equals the product of the area of the pre-image and the square of the measure of the scale factor.

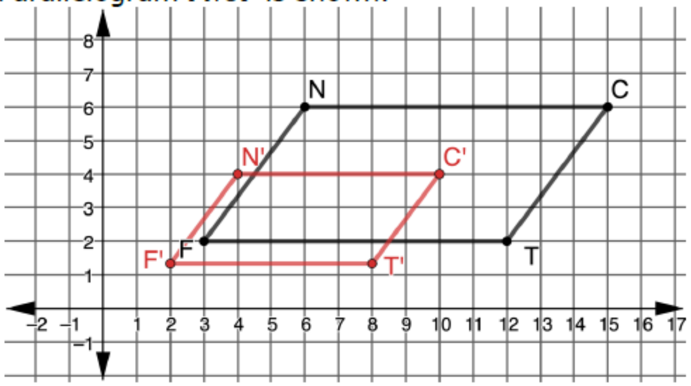
Figure	Length	Width	Area
Pre-Image	3	7	21 $units^2$
Image	9	21	189 $units^2$

Allow students to use colored pencils and/or pens to graph on the coordinate plane to make the graphs more visual.

Consider asking the following question to promote student discussion:

- *Why does the conjecture about the relationship between area and scale factor for the square not hold true for the rectangle?*

5. Parallelogram $FNCT$ is shown.



a. What is the area of $FNCT$, in square units?
The area of $FNCT$ is $36\ units^2$.

b. Dilate $FNCT$ by a scale factor of $\frac{2}{3}$ centered at the origin. Write the coordinates of the preimage and image in the table.

Point	Preimage	Image
F	$(3,2)$	$\left(2,\frac{4}{3}\right)$
N	$(6,6)$	$(4,4)$
C	$(15,6)$	$(10,4)$
T	$(12,2)$	$\left(8,\frac{4}{3}\right)$

c. Plot the coordinates of $F'N'C'T'$ on the same coordinate plane as $FNCT$.

d. What is the relationship between the area of $FNCT$ and $F'N'C'T'$?
The area of $F'N'C'T'$ is $16\ unit^2$, which is $\frac{4}{9}$ the area of $FNCT$. The value $\frac{4}{9}$ is the square of the scale factor, $\frac{2}{3}$.

Before having students fill in the table to find the coordinates of the image, have students make a prediction about what they think the area of the image will be when compared to the preimage.

9.10.2 Exploration: Scale Factor and Area (continued)

e. How can the scale factor of $\frac{2}{3}$ be used to determine the area of $F'N'C'T'$ from the area of $FNCT$?
Since each dimension is $\frac{2}{3}$ of the original figure, the area of parallelogram $F'N'C'T'$ will be $(\frac{2}{3} \times base)(\frac{2}{3} \times height)$, which means that the area of $F'N'C'T'. = \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)(area\ of FNCT) = \left(\frac{4}{9}\right)(area\ of FNCT)$.

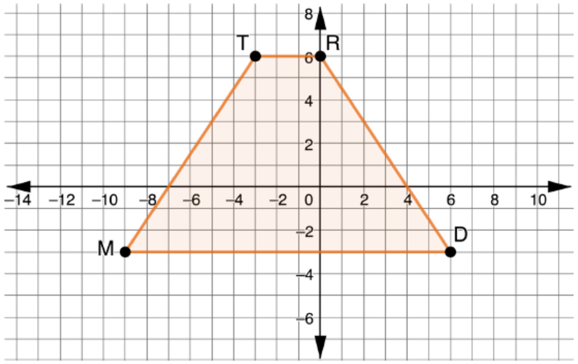
- *Does the conjecture for the rectangle hold true for the parallelogram? (Yes.)*
- *How do you calculate the ratio of the areas of parallelogram $FNCT$ and parallelogram $F'N'C'T'$? (Divide the area of the preimage by the area of the image.)*

9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area

Component Overview: *After 9.10.2 Exploration, students will apply the skills and conjectures to find the area of a trapezoid preimage and then perform dilations to find the area and determine the effects that the scale factor has on the area.*

9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area

1. Trapezoid $MTRD$ is shown on the coordinate plane.



a. Find the area of $MTRD$.

$$A = \frac{1}{2}(15 + 3)(9)$$

The area of $MTRD$ is 81 square units.

b. Find the coordinates of $MTRD$ given the scale factors in the table. Use the preimage coordinates to find the coordinates of the images. Both dilations are centered at $(0,0)$.

Point	Preimage	Dilation $k = \frac{1}{6}$ ($M'T'R'D'$)	Dilation $k = \frac{4}{3}$ ($M"T"R"D"$)
M	$(-9,-3)$	$\left(-\frac{3}{2},-\frac{1}{2}\right)$	$(-12,-4)$
T	$(-3,6)$	$\left(-\frac{1}{2},1\right)$	$(-4,8)$
R	$(0,6)$	$(0,1)$	$(0,8)$
D	$(6,-3)$	$\left(1,-\frac{1}{2}\right)$	$(8,-4)$

What is something you notice about this trapezoid? What is something that you wonder?

Allow students who need extra support with fractions to use a calculator.

9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area (continued)

c. Find the area of $M'T'R'D'$.

$$A = \frac{1}{2}(2.5 + 0.5)(1.5)$$

The area of trapezoid $M'T'R'D'$ is $2.25\,\text{unit}^2$.

d. What effects does the scale factor of $\frac{1}{6}$ have on the original area?
The product of the square factor and the area of the pre-image is equal to the area of the image.
$\left(\frac{1}{6}\right)^2 \cdot 81 = 2.25\ unit^2$

e. Find the area of $M''T''R''D''$.
$A = \frac{1}{2}(20 + 4)(12)$

The area of $M''T''R''D''$ is $144\ \text{unit}^2$.

f. What effects does the scale factor of $\frac{4}{3}$ have on the original area?
The product of the square factor and the area of the pre-image is equal to the area of the image.
$\left(\frac{4}{3}\right)^2 \cdot 81 = 144\,unit^2$

g. How does the area of the trapezoid with a scale factor of $\frac{1}{6}$ differ from the area of the trapezoid with a scale factor of $\frac{4}{3}$?
Answers will vary. The area of a trapezoid with a scale factor of $\frac{1}{6}$ will be much smaller than the area of the original trapezoid. The area of a trapezoid with a scale factor of $\frac{4}{3}$ will be larger than the area of the original trapezoid.

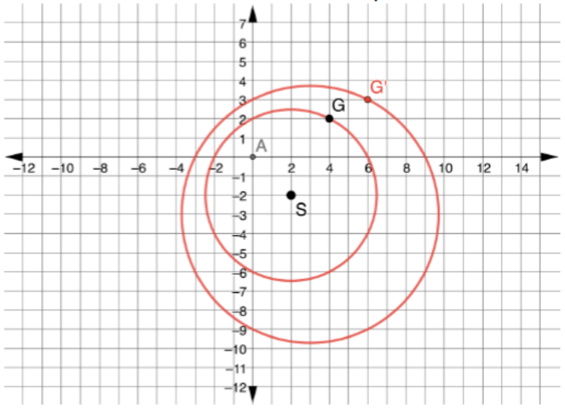
Allow students to discuss questions 1d, 1f, and 1g with a partner or whole group. Consider having students use the Literacy and Language Routine Take Turns when sharing their responses to the questions.

9.10.4 Your Turn!

Component Overview: *Students are given a circle with a radius that is a square root. Students graph the image and find the areas of the preimage and the image.*

9.10.4 Your Turn!

1. Circle S is shown with a radius of $\sqrt{20}$.



a. Dilate circle S using a scale factor of 1.5 centered at the origin, and graph the resulting image on the coordinate plane.
See the graph.

b. Find the area of the circles.

Area of Preimage	Area of Image
$r = \sqrt{20}$ $A = \pi r^2$ $A = \pi\left(\sqrt{20}\right)^2$ $A = 20\pi$	$r' = \sqrt{45}$ $A = \pi(r'^2)$ $A = \pi\left(\sqrt{45}\right)^2$ $A = 45\pi$

When students are finding the radius of the image, students may want to multiply 20 by 1.5, which would give students 30, but this will yield an incorrect radius. Students must multiply $(\sqrt{20})(1.5)$, which equals $\sqrt{45}$.

Remind students of the formula of the area of the circle if necessary.

For students who finish early, challenge them to create a third circle with another dilation. Have students then calculate the area and the radius of the circle and compare.

9.10.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

9.10.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about how dilations affect area.

   Answers will vary.

9.11 Determining Surface Area Given Dimension Changes

Lesson Introduction

Benchmark	MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the areas of two-dimensional figures and the surface area or volume of three-dimensional figures.	Learning Targets	• I can find the area of a figure when one or both dimensions are changed. • I can find the surface area of a three-dimensional figure when one or both dimensions are changed.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions

- How can you find the area of a figure when one or both dimensions are changed?
- How do you find the surface area of a three-dimensional figure when one or both dimensions are changed?
- What effect(s) does changing the dimensions have on the area or surface area of a figure?

Lesson Narrative

The lesson guides students to notice the impacts that changing dimensions have on area and surface area and provides several real-world scenarios that students work through and discuss with classmates. The lesson builds on previous lessons on area, volume, and surface area to discover the impact that dimension changes have on the area, volume, and surface area.

Component Summary

9.11.1 Warm-Up (~5 min.)	Determining the area of a trapezoid and the coordinates of a dilated image *(SE p. 112)*	9.11.3 Guided Practice (10 min.)	Finding area and surface area when dimensions are changed *(SE p. 115)*
9.11.2 Exploration (15-20 min.)	Evaluating how changing dimensions impacts area and cost of a home project *(SE p. 113)*	9.11.4 Your Turn! (5-10 min.)	Determining how the area changes when dimensions are changed *(SE p. 116)*
9.11.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 116)*		

Resources

Glossary	Materials	Additional Preparation
<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Scale factor • Dilation	• (Optional) Chart Paper for Students to Record Results • (Optional) Colored Pens/Pencils • (Optional) Graphing Utility	• If necessary, find a GeoGebra applet using the phrase “Scale Factor and Area” and “Scale Factor and Volume.”

	Common Misconceptions	Things to Consider
Teacher Tips	• On question 7 in 9.11.2 Exploration, students may forget to subtract the part of the surface that is water when calculating the surface area (see notes in lesson guide). • Students may not apply the correct formulas for area, volume, and/or surface area.	• Consider having students record their results in groups using chart paper or other forms of recording data.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect This lesson provides an opportunity for students to connect and compare their learning to a real-world scenario. Provide opportunities for students to reflect on their learning throughout the lesson.	**MA.K12.MTR.7.1: Real-World Contexts** Lesson 11 is written purposefully using real-world concepts that have students connect mathematics to everyday experiences. Allow students to work through the questions in the lesson and use various methods to solve.
Differentiate Instruction	Consider making the 9.11.2 Collaboration, 9.11.3 Guided Practice, and 9.11.4 Your Turn! into separate stations that students can work through. • Station 1: Collaboration (questions 1-3) • Station 2: Collaboration (questions 4-6) • Station 3: Collaboration (question 7-9); be sure that students have completed Stations 1 and 2 first. Station 3 may also be a teacher-centered station. • Station 4: Guided Practice • Station 5: Your Turn!	

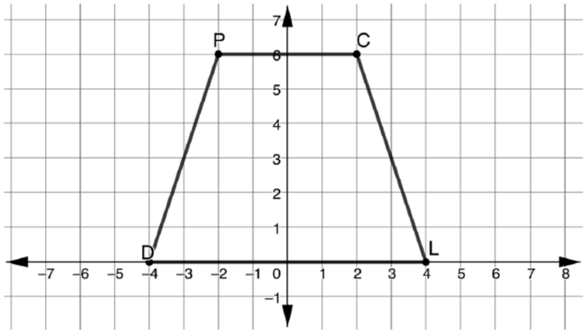
Lesson Notes

9.11.1 Warm-Up: Bell Work

Component Overview: *Students find the area of the given trapezoid and then perform a dilation with a given scale factor. This ties into Lesson 10.*

9.11.1 Warm-Up: Bell Work

1. Trapezoid $DPCL$ is shown in the coordinate plane.



a. Find the area of $DPCL$.

$A = \frac{1}{2}(8+4)(6)$
$A = 36\ unit^2$

b. Find the coordinates of the image of $MTRD$ given the scale factor in the table. Use the preimage coordinates to find the coordinates of the image. The dilation is centered at $(0,0)$.

Point	Preimage	Dilation $k = 6$ $(D'P'C'L')$
D	$(-4,0)$	$(-24,0)$
P	$(-2,6)$	$(-12,36)$
C	$(2,6)$	$(12,36)$
L	$(4,0)$	$(24,0)$

Can you find the area of the trapezoid by dividing into triangles and rectangles?

Consider allowing students to use either the formula for the area of a trapezoid or divide into triangles and a rectangle.

9.11.2 Exploration: Ryan's Renovations

Component Overview: *9.11.2 Exploration guides students through scenarios that would be used in home renovations, thus providing a real-world context. Students find the area of a figure and apply new dimensions through changing the sizes of a kitchen and a swimming pool.*

9.11.2 Exploration: Ryan's Renovations

Ryan is building a new house and wants to make the size of the kitchen larger. The current kitchen is 24 feet wide and 10 feet long. Ryan wants to make the new kitchen 20 feet long but leave the width the same.

1. Find the area of the original kitchen and the area of the new kitchen.

Area of the Original Kitchen	Area of the New Kitchen
$A = 10 \cdot 24$ $A = 240\ ft^2$	$A = 20 \cdot 24$ $A = 480\ ft^2$

2. How do the areas compare?
   The area of the new kitchen is twice the area of the original kitchen.

Ryan also wants to install a pool for his new house. The construction company gives Ryan three shape options for the new pool. Each pool's dimensions are given in feet (ft).

Pool 1	Pool 2	Pool 3
Cylindrical	**Rectangular Prism**	**Cube**
diameter = 16 ft height= 12 ft	length= 30 ft width= 20 ft height= 12 ft	Each side= 15 ft

3. Find the surface area of each of the shapes for the three pool options. Round to the nearest thousandth.

Pool 1	Pool 2	Pool 3
$SA = 2B + Ch$ $SA = 128\pi + 192\pi$ $SA = 320\pi$ $SA \approx 1{,}005.310\ ft^2$	$SA = 2B + Ph$ $SA = 1{,}200 + 1{,}200$ $SA = 2{,}400\ ft^2$	$SA = 6s^2$ $SA = 6(225)$ $SA = 1{,}350\ ft^2$

Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions and which students help clarify.

Consider using 9.11.2 Exploration as three separate stations (see Differentiate Instruction notes on page 2 of the lesson guide).

The collaboration gives an opportunity to fit in MA.K12.MTR.7.1: Real-World Contexts, where students are applying mathematics. to real-world contexts.

On Pool 1, have students divide the diameter by 2 to find the radius.

The part of the surface area that is water is included in these calculations. That part of the surface area will be subtracted later in the lesson.

9.11.2 Exploration: Ryan's Renovations (continued)

The construction company is giving Ryan more options for building a swimming pool. The three options are listed in the table.

Pool 4	Pool 5	Pool 6
Cylindrical	**Rectangular Prism**	**Cube**
diameter $= 8$ ft height $= 6$ ft	length $= 15$ ft width $= 10$ ft height $= 6$ ft	Each side $= 7.5$ ft

4. Describe how the dimensions of pair of pools are related.
   Answers will vary.

Pair	Description
Pool 1 and Pool 4	*The dimensions of Pool 1 are twice as large as the dimensions of Pool 4.*
Pool 2 and Pool 5	*The dimensions of Pool 2 are twice as large as the dimensions of Pool 5.*
Pool 3 and Pool 6	*The dimensions of Pool 3 are twice as large as the dimensions of Pool 6.*

5. Find the surface area of each of the shapes for the three new options for a pool. Round to the nearest thousandth.

Pool 4	Pool 5	Pool 6
$SA = 2B + Ch$ $SA = 32\pi + 48\pi$ $SA = 80\pi$ $SA \approx 251.327\ ft^2$	$SA = 2B + Ph$ $SA = 300 + 300$ $SA = 600\ ft^2$	$SA = 6s^2$ $SA = 6(56.25)$ $SA = 337.5\ ft^2$

Consider opportunities to have students display their work in the classroom when working through the collaboration. Chart or regular paper can be used for each pool.

How did the areas of Pools 4, 5, and 6 differ from Pools 1, 2, and 3? (The surface areas of Pools 4, 5, and 6 are $\frac{1}{4}$ of the surface areas of Pools 1, 2, and 3.)

The part of the surface area that is water is included in these calculations. That part of the surface area will be subtracted later in the lesson.

9.11.2 Exploration: Ryan's Renovations (continued)

6. Record the surface area of all pool shapes in the table below.

Pool	Surface Area
1	$SA \approx 1{,}005.31\ ft^2$
2	$SA = 2{,}400\ ft^2$
3	$SA = 1{,}350\ ft^2$
4	$SA \approx 251.327\ ft^2$
5	$SA = 600\ ft^2$
6	$SA = 337.5\ ft^2$

The cost of building the swimming pool is dependent upon the amount of material. Since the top of each of the shapes would be water, determine the surface area of each pool that could be used to determine the cost.

7. Complete the table.

Pool	Surface Area for Material
1	$SA = 804.248\ ft^2$
2	$SA = 1{,}800\ ft^2$
3	$SA = 1{,}125\ ft^2$
4	$SA = 201.062\ ft^2$
5	$SA = 450\ ft^2$
6	$SA = 281.25\ ft^2$

8. If Ryan wants the least expensive pool, which one should he purchase?
   Ryan should purchase Pool 4 because it has the smallest surface area.

9. Which pool would you purchase? State your reasoning.
   Answers will vary.
   Pool 2 because it will give more space in which to swim.

Students must subtract the surface that is the water in question 7. Monitor students' work to show that this is reflected in their final answers.

Consider providing students an opportunity to discuss their personal choices and reasons for which pool they would purchase.

9.11.3 Guided Practice: Ryan's House

Component Overview: *9.11.3 Guided Practice gives a scenario about the installation of a shower and a bathtub. Students find the surface area to determine whether the shower or tub should be installed.*

9.11.3 Guided Practice: Ryan's House

Ryan is building a bathroom and needs to install either a bathtub or a shower. The options for each bathroom size are given.

Shower 1	Tub 1
length $= 4$ ft width $= 6$ ft height $= 12$ ft	length $= 4$ ft width $= 8$ ft height $= 10$ ft

1. In order to determine the amount of drywall, Ryan needs to find the surface area of each of the room options.

Shower 1	Tub 1
$SA = B + Ph$ $SA = 24 + 240$ $SA = 264\ ft^2$	*Calculating SA without the top.* $SA = B + Ph$ $SA = 32 + 240$ $SA = 272\ ft^2$

2. Ryan determines that the dimensions of the shower and tub are not up to code. The updated dimensions are shown. Use the new dimensions to determine the surface area of each room.

Shower 2	Tub 2
length $= 6$ ft width $= 9$ ft height $= 12$ ft	length $= 6$ ft width $= 12$ ft height $= 15$ ft
Surface Area	**Surface Area**
$SA = B + Ph$ $SA = 54 + 360$ $SA = 414\ ft^2$	$SA = B + Ph$ $SA = 72 + 540$ $SA = 612\ ft^2$

Continue highlighting MA.K12.MTR.7.1: Real-World Contexts in 9.11.3 Guided Practice.

Make sure that students do not find the surface area of the top of the tub.

9.11.3 Guided Practice: Ryan's House (continued)

Two of Shower 2's dimensions were changed by a factor of 1.5, whereas all of Tub 2's dimensions were changed by a factor of 1.5. Discuss with your partner on how to use the factor 1.5 and the number of changes in dimensions to find the new surface area from the original surface area.

3. Complete the statements.

$\underline{1.568}$ × surface area of Shower 1 = surface area of Shower 2

$\underline{2.25}$ × surface area of Tub 1 = surface area of Tub 2

Students are not required to write a summary of their discussions. Listen as students discuss their thinking in response to this prompt. Student thinking should highlight that the scale factor is related to the area if the scale factor is applied to all dimensions of the shape but not just some dimensions.

Question 3 is algebraic equations that are solved by substituting in the values from the tables above.

9.11.4 Your Turn!

Component Overview: *Students work through a scenario involving a living room and are asked to find the area and analyze the effect of a dimension change with a given scale factor.*

9.11.4 Your Turn!

The living room of Ryan's house is 12 feet long and 18 feet wide.

1. The room is a rectangular shape. Find the area of Ryan's living room.
   $A = 216\ ft^2$

2. How would the area of the living room change if Ryan decided to enlarge the dimensions by 25%?

Old Dimensions	*New Dimensions*
Length $= 12\ ft$ *Width* $= 18\ ft$	*Length* $= 15\ ft$ *Width* $= 22.5\ ft$

   New area $= 337.5\ ft^2$. *This is an enlargement.*

3. What is the scale factor for a dimension increase of 25%?
   The scale factor is 1.25.

4. How did the area of the living room change with the new dimensions?
   $1.25^2 = 1.5625$
   The area of the living room increased by 56.25%

Continue highlighting MA.K12.MTR.7.1: Real-World Contexts in 9.11.4 Your Turn!

On question 3, students may forget to add 100% *to the* 25% *to find* 1.25 *as the scale factor.*

For students who finish early, challenge them to create dimensions for another room and how the area would change with new dimensions based on a scale factor for another room in Ryan's house.

9.11.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

9.11.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

   Answers will vary.